Closure Models, Physics-Informed Neural Networks, Neural Operators, and World Models

A unified numerical-approximation perspective

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Abstract

This chapter develops a common language for several ideas that are often discussed in separate communities: closure modeling in fluid dynamics and kinetic theory, physics-informed neural networks, neural operators, structure-preserving scientific machine learning, and world models. The central observation is deliberately simple. A neural network is a flexible numerical approximation family; the decisive question is *which mathematical object it is approximating*. It may approximate one solution of a differential equation, an input-to-solution operator, a constitutive or closure relation, a dynamical vector field, a time-advance map, an invariant manifold, or an action-conditioned transition kernel. Once that object is identified, the roles of equations, data, numerical solvers, physical constraints, and recursive prediction become much clearer.

Particular attention is given to four complementary research programs. Work by Zaher Hani and collaborators on long-time particle-to-kinetic, kinetic-to-fluid, and wave-to-kinetic limits supplies canonical physical settings in which closure, factorization, cumulants, and interaction history can be tested against a known asymptotic answer. A substantial survey of machine learning for *classical, non-quantum many-body systems* connects statistically identifiable interaction-kernel inference, geometry-aware and graph-based particle models, weak-form equation discovery, coarse-grained dynamics, and experiment-facing force decomposition. The dusty-plasma case provides a concrete bridge from particle trajectories to a TEMPEST application profile. Work by Andrew Christlieb and collaborators develops hyperbolic machine-learned kinetic moment closures, while Qi Tang and collaborators study stabilized neural ODEs, fast-slow neural networks, symplectic neural maps, energetic variational learning, and local-global neural operators. These methods are compared with classical residual PINNs and with neural operators in the standard function-space sense. The final sections explain how the same approximation viewpoint extends naturally to world models: learned internal simulators that estimate latent state, predict action-conditioned futures, and support planning or policy learning.

Contents

1 The organizing idea: what mathematical object is the network approximating?

At an extremely high level, a neural network is a flexible, parameterized family of functions. Training chooses parameters θ so that one member of that family approximates a desired mathematical object. The object may be a scalar-valued function, a vector field, an operator between function spaces, a constitutive relation, a coordinate transformation, or a probability kernel.

The vocabulary changes from field to field because the *role* of the approximation changes. The same broad mathematical technology can therefore appear under very different names.

The gross-level unification

$$\boxed{\text{neural network} = \text{a flexible numerical ansatz for a function, map, operator, or kernel.}}$$

The scientifically meaningful questions are:

1. What object is being approximated?
2. What information is used to fit it: equations, data, or both?
3. Which mathematical properties are merely encouraged, and which are guaranteed by construction?
4. Does the network replace the whole numerical solver, or only one component of it?
5. Will the learned object be evaluated once, integrated in time, or composed recursively for long rollouts?

A useful first taxonomy is

$$\begin{aligned}
u_\theta(x,t) &\approx u(x,t), && \text{a particular solution,}\\
\mathcal{G}_\theta(a) &\approx \mathcal{G}^\dagger(a), && \text{an input-to-solution operator,}\\
\mathcal{C}_\theta(m,\nabla m) &\approx \text{an unresolved closure term,}\\
F_\theta(z) &\approx F(z), && \text{a dynamical vector field,}\\
\Phi_\theta(z,a) &\approx \Phi(z,a), && \text{a controlled time-advance map,}\\
h_\theta(x) &\approx h(x), && \text{an invariant or slow manifold,}\\
P_\theta(\,\mathrm{d}z' \mid z,a) &\approx P(\,\mathrm{d}z' \mid z,a), && \text{a stochastic transition kernel.}
\end{aligned} \tag{1}$$

This classification already separates the major paradigms:

Paradigm	Typical learned object	Numerical role
Classical PINN	One function $u_\theta(x,t)$	Neural trial function for one forward or inverse problem
Neural operator	Map $a(\cdot) \mapsto u(\cdot)$	Amortized solver for a family of problem instances
Learned closure	Missing relation $\mathcal{C}_\theta$	Component inserted into a retained reduced PDE or ODE model
Neural ODE	Vector field F_θ	Learned right-hand side integrated by an ODE solver
Learned flow map	Step map Φ_θ	Direct state-to-state surrogate, often recursively composed
World model	Latent state model and transition kernel	Internal simulator for prediction, counterfactual rollout, and planning

The rest of the chapter makes these distinctions precise, beginning with the classical closure problem in fluid dynamics.

2 Closure models and closure theories in fluid dynamics

2.1 Closure begins before turbulence

Fluid mechanics is built from balance laws. Conservation of mass, momentum, and energy constrains the evolution of density, velocity, and internal energy, but these balances do not by themselves specify the material response. A momentum equation contains a stress tensor; an energy equation contains a heat flux. To obtain a usable continuum model one introduces constitutive relations, such as

$$\boldsymbol{\sigma} = -pI + 2\mu S + \lambda(\nabla \cdot u)I, \qquad q = -\kappa \nabla T,$$

together with an equation of state. The Newtonian stress law, Fourier heat conduction, and the equation of state are already *closures*. They replace microscopic information by relations among retained macroscopic variables.

Thus the Navier–Stokes–Fourier equations are not raw conservation laws. They are a closed continuum model of a more microscopic kinetic system.

2.2 The general closure mechanism

Suppose a full state $X(t)$ evolves by

$$\dot{X} = \mathcal{F}(X),$$

but we retain only reduced variables

$$a = \Pi X.$$

Projecting or averaging the full equations often gives an exact identity for a, but the identity contains quantities depending on the discarded part of X. A closure replaces those unresolved quantities by a model involving the retained variables, perhaps including gradients, history, stochastic forcing, or auxiliary variables.

Exact reduced equation versus approximate closure

Averaging, filtering, taking moments, or projecting modes can produce an exact reduced equation. The approximation enters when unresolved correlations, fluxes, memory terms, or conditional expectations are replaced by a tractable model.

This distinction is essential. The reduced equation and the closure are conceptually different objects, even when they are written together as one model.

2.3 Reynolds-averaged turbulence and the moment hierarchy

For incompressible Navier–Stokes,

$$\partial_t u_i + u_j \partial_j u_i = -\partial_i p + \nu \Delta u_i, \qquad \partial_i u_i = 0, \tag{2}$$

write

$$u_i = U_i + u_i', \qquad \langle u_i' \rangle = 0.$$

Averaging gives

$$\partial_t U_i + U_j \partial_j U_i = -\partial_i \bar{p} + \nu \Delta U_i - \partial_j R_{ij}, \tag{3}$$

where

$$R_{ij} = \langle u_i' u_j' \rangle$$

is the Reynolds-stress tensor. The equation for the mean velocity is not closed because R is not determined by U.

One can derive a transport equation for R_{ij}. Schematically,

$$\frac{DR_{ij}}{Dt} = P_{ij} + \Phi_{ij} - \varepsilon_{ij} + D_{ij}, \tag{4}$$

where production is largely determined by resolved mean gradients, while pressure–strain redistribution, dissipation, and turbulent transport involve new correlations. Third moments such as

$$\langle u_i' u_j' u_k' \rangle$$

appear; their equations involve fourth moments; and so on. This is the turbulence moment hierarchy.

2.4 RANS closures

Reynolds-averaged Navier–Stokes models seek mean quantities without resolving individual turbulent eddies (Pope, 2000).

2.4.1 Linear eddy viscosity

The classical Boussinesq form is

$$R_{ij} = \frac{2}{3}k\delta_{ij} - 2\nu_t S_{ij}, \qquad k = \frac{1}{2}R_{ii}, \tag{5}$$

with

$$S_{ij} = \frac{1}{2}(\partial_j U_i + \partial_i U_j).$$

The eddy viscosity ν_t models turbulent momentum transport. The key restriction is tensorial: the anisotropic stress is assumed locally aligned with the mean strain.

A mixing-length closure may use

$$\nu_t = \ell_m^2 \left| \frac{\mathrm{d}U}{\mathrm{d}y} \right|.$$

One-equation models transport a viscosity-like or energy-like variable. Two-equation models transport two turbulent quantities, for example

$$\nu_t = C_\mu \frac{k^2}{\varepsilon} \qquad \text{or} \qquad \nu_t \sim \frac{k}{\omega}.$$

The extra equations do not eliminate closure; their production, diffusion, dissipation, and wall behavior must themselves be modeled.

2.4.2 Nonlinear algebraic stress and Reynolds-stress transport

A nonlinear eddy-viscosity model permits a richer tensor relation

$$b = \frac{R}{2k} - \frac{1}{3}I = \sum_n g_n(I_1, I_2, \ldots) T^{(n)}(S, \Omega), \tag{6}$$

where $T^{(n)}$ are invariant tensor-basis elements formed from strain S and rotation Ω.

A Reynolds-stress transport model instead evolves the six independent entries of R, plus one or more scale variables. This gives more freedom for anisotropy, rotation, curvature, and secondary flows, but the pressure–strain, dissipation, and triple-correlation transport terms remain unclosed. Closure has moved deeper into the hierarchy rather than disappeared.

2.5 Large-eddy simulation closure

Large-eddy simulation filters the velocity rather than taking a long-time or ensemble mean. Filtering gives

$$\partial_t \overline{u}_i + \overline{u}_j \partial_j \overline{u}_i = -\partial_i \overline{p} + \nu \Delta \overline{u}_i - \partial_j \tau_{ij}, \tag{7}$$

where

$$\tau_{ij} = \overline{u_i u_j} - \overline{u}_i\, \overline{u}_j \tag{8}$$

is the subgrid-scale stress. LES resolves large eddies and models the transfer across a chosen filter or grid scale.

2.5.1 Functional closures

A functional model emphasizes the net transfer of energy between resolved and unresolved scales. The Smagorinsky model uses

$$\tau_{ij}^{d} = -2\nu_{\mathrm{sgs}}\overline{S}_{ij}, \qquad \nu_{\mathrm{sgs}} = (C_s\Delta)^2|\overline{S}|, \tag{9}$$

following the subgrid-viscosity idea of Smagorinsky (1963). Its modeled energy transfer is

$$\Pi = -\tau_{ij}\overline{S}_{ij} = 2\nu_{\mathrm{sgs}}\overline{S}_{ij}\overline{S}_{ij} \geq 0, \tag{10}$$

so the model always removes energy from the resolved field. This is stabilizing and represents the average forward cascade, but it excludes local backscatter.

Dynamic models infer coefficients from the resolved flow by comparing two filter levels, using the Germano identity (Germano et al., 1991).

2.5.2 Structural closures and implicit LES

Structural models seek to approximate the tensor τ itself. Scale-similarity models, gradient expansions, approximate deconvolution, and mixed models fall into this category. They can represent stress structure and backscatter more faithfully, but they may supply too little net dissipation for a stable computation.

In implicit LES no explicit subgrid stress is written. Dissipation from reconstruction, limiting, fluxes, and time stepping acts as an effective closure. The numerical discretization is then part of the physical model rather than a neutral implementation detail.

2.6 Statistical closure theories

In homogeneous turbulence one can derive evolution equations for two-point correlations or spectra. A two-point covariance

$$C_{ij}(x-y,t) = \langle u_i(x,t)u_j(y,t)\rangle$$

or an energy spectrum $E(k,t)$ evolves through third-order correlations representing triadic interactions. Equations for third-order correlations involve fourth-order correlations.

Classical closure strategies include quasi-normal approximations, eddy-damped quasi-normal models, direct-interaction approximation, and Lagrangian-history variants. Kraichnan's direct-interaction program introduced response functions alongside correlations and derived self-consistent spectral equations (Kraichnan, 1959).

An essential requirement is *realizability*. A modeled covariance must remain positive semidefinite, and an energy spectrum cannot become negative. A statistical closure that violates these conditions is not merely inaccurate; it no longer corresponds to any probability law.

2.7 Kinetic and moment closure

Let $f(x,v,t)$ be a kinetic distribution. Its moments are

$$M^{(n)}(x,t) = \int v^{\otimes n} f(x,v,t)\,\mathrm{d}v. \tag{11}$$

The equation for the n-th moment generally contains the $(n+1)$-st moment. Retaining density, momentum, and energy does not automatically determine stress and heat flux.

Different assumptions on f produce different fluid closures:

- a local Maxwellian gives compressible Euler;
- near-equilibrium asymptotics give Navier–Stokes–Fourier;
- Grad systems retain finitely many polynomial or Hermite moments;
- Gaussian closures retain anisotropic covariance information;
- maximum-entropy closures choose a distribution by constrained entropy optimization.

Entropy-based moment hierarchies make structural goals such as hyperbolicity, realizability, and entropy dissipation explicit (Levermore, 1996).

2.8 Mori–Zwanzig: exact closure generally contains memory

Projection-operator theory gives a particularly clean abstract account. If the full dynamics contain resolved variables a and unresolved variables b, then eliminating b leads formally to a generalized Langevin equation

$$\dot{a}(t) = \mathcal{M}(a(t)) + \int_0^t \mathcal{K}(a(t-s), s)\, \mathrm{d}s + \eta(t). \tag{12}$$

The three pieces are an instantaneous term, a memory term, and an orthogonal-dynamics or fluctuating term (Mori, 1965; Zwanzig, 1961).

A crucial closure lesson

Exact reduced dynamics are generally nonlocal in time and depend on unresolved initial information. A local deterministic closure

$$\dot{a} = F(a)$$

is therefore a modeling assumption, not a generic consequence of eliminating variables.

This point will return later. A neural ODE trained on coarse observations may not mention the original PDE, yet it is implicitly making a Markovian closure assumption.

2.9 Data-driven closure

Machine learning changes the representation or calibration of the closure; it does not remove the closure problem. One may learn

$$R_{ij} = \mathcal{C}_{ij}(U, \nabla U, k, \varepsilon, \ldots),$$

a subgrid stress, a correction to a production term, an unresolved kinetic moment, or a memory kernel.

A credible learned closure should respect as much known structure as possible:

$$\begin{gathered}\text{frame invariance,} \quad \text{tensor symmetry,} \quad \text{conservation,} \quad \text{realizability,} \\ \text{correct asymptotic limits,} \quad \text{stable energy behavior.}\end{gathered} \tag{13}$$

It is also important to distinguish two tests:

a priori: Does the closure match high-fidelity targets on stored data?

a posteriori: Does the coupled reduced equation plus closure produce the correct evolution?

A highly accurate offline stress fit can still destabilize the closed PDE, alter its attractor, or feed itself inputs outside the training distribution.

2.10 What makes a closure credible?

Across RANS, LES, kinetic moments, and data-driven reduction, several requirements recur.

1. **Symmetry and invariance.** The model should transform correctly under rotations, translations, and changes of inertial frame.
2. **Realizability.** For example, a Reynolds stress satisfies

$$R = \langle u' u'^T \rangle \succeq 0, \qquad |R_{ij}|^2 \leq R_{ii} R_{jj}.$$

3. **Correct energy or entropy behavior.** The closure must put energy in the right scales and must not generate forbidden growth.
4. **Correct limits.** It should recover laminar, near-equilibrium, diffusive, near-wall, or mesh-refined limits where appropriate.
5. **Nonlocality and memory.** A purely local instantaneous closure is a substantive simplification.
6. **Well-posedness.** A plausible stress formula is not enough if the resulting PDE loses hyperbolicity, parabolicity, or numerical stability.

The next sections explain how neural approximations enter these problems, and why different scientific-machine-learning paradigms learn different mathematical objects.

3 Zaher Hani and collaborators: rigorous scale limits as canonical closure benchmarks

The work of Zaher Hani and collaborators is not a catalog of plug-in closure formulas. It is more foundational: it proves, in carefully controlled physical regimes, when a microscopic or wave description reduces to a kinetic equation and when a kinetic equation reduces further to fluid mechanics. Those results identify the scaling limit, the information discarded by the reduction, the time horizon over which the reduction remains valid, and the mathematical structures used to control the discarded information. For a computational closure program, that combination is unusually valuable.

The three relevant papers form a connected chain. Deng, Hani, and Ma derive the Boltzmann equation from a rarefied hard-sphere gas for arbitrarily long times within the lifespan of the Boltzmann solution (Deng et al., 2025a). A companion paper connects hard-sphere dynamics, through Boltzmann kinetic theory, to compressible Euler and incompressible Navier–Stokes–Fourier limits (Deng et al., 2025b). Deng and Hani obtain a long-time justification of the homogeneous wave kinetic equation over its full lifespan in a weakly nonlinear random-wave regime (Deng and Hani, 2024).

3.1 What the three results contribute

3.1.1 Hard spheres to Boltzmann

The microscopic system consists of elastically colliding hard particles in a rarefied-gas scaling. The reduced description is the one-particle distribution governed by the Boltzmann equation. The central closure assumption is propagation of chaos: finite collections of particles become approximately independent in the kinetic limit, so that the collision term can be written using products of one-particle distributions.

At finite density and finite scale separation, independence is not exact. Recollisions and shared collision ancestry produce connected correlations. The long-time proof propagates a structured cumulant ansatz, restarts partial expansions across time layers, and organizes collision histories through diagrams called *molecules*. Computationally, these objects suggest measurable diagnostics: projected two- and three-particle cumulants, recollision counts, collision-graph connectivity, factorization error, and the residual of the Boltzmann equation.

3.1.2 Boltzmann to fluid

The companion result continues the reduction from kinetic theory to classical fluid equations. Compressible Euler arises from a local-equilibrium regime, while incompressible Navier–Stokes–Fourier arises under a different hydrodynamic scaling and preparation of the kinetic data. This makes the work especially useful for closure research because it displays two successive information losses:

$$\text{Newtonian particles} \longrightarrow \text{Boltzmann distribution} \longrightarrow \text{fluid fields and constitutive fluxes.}$$

The first step suppresses particle correlations; the second suppresses velocity-space information beyond a few hydrodynamic moments. A numerical benchmark can therefore test not only whether the end-to-end fluid limit is recovered, but also which error enters at each reduction and whether a learned correction is attached to the correct level of the hierarchy.

3.1.3 Nonlinear waves to wave kinetics

The wave-turbulence result treats a different microscopic mechanism but the same closure logic. A nonlinear dispersive wave field has interacting Fourier modes; the reduced wave kinetic equation evolves a spectrum rather than the phases and higher joint statistics of those modes. Random-phase or near-Gaussian factorization is the analogue of molecular chaos. Connected spectral cumulants and resonant interaction histories measure the information discarded by that closure.

The long-time analysis again uses time layering and structured interaction diagrams, here called canonical layered gardens. This supplies a second canonical testbed in which the unresolved information is spectral rather than particle based. Agreement of a closure diagnostic across the particle and wave tracks would therefore be stronger evidence than success on two variants of the same simulator.

3.2 Why these are unusually good benchmark settings

These regimes are "standard" in a precise sense: hard-sphere kinetic theory, Boltzmann hydrodynamic limits, and weakly nonlinear wave kinetics are canonical models with explicit microscopic dynamics, explicit reduced equations, and named small parameters. They are substantially cleaner than a driven, bounded, magnetized, multispecies TEMPEST application, but that simplicity is an advantage at the validation stage.

Canonical reduction	**Classical closure**	**Discarded information**	**Benchmark evidence**
Hard particles → Boltzmann	Molecular chaos / factorization	Connected particle cumulants and collision ancestry	Convergence of marginals, collision statistics, cumulants, and Boltzmann residuals
Boltzmann → fluid	Local equilibrium and constitutive asymptotics	Non-Maxwellian velocity structure, higher moments, and kinetic remainder	Recovery of Euler or Navier–Stokes–Fourier fields, stresses, heat fluxes, and scaling rates
Nonlinear waves → wave kinetics	Random-phase or near-Gaussian spectral factorization	Spectral cumulants, phase information, and resonant interaction history	Convergence of spectra, transfer rates, spectral cumulants, and wave-kinetic residuals

They provide six properties that an evaluation harness needs:

1. **A known target.** The limiting kinetic or fluid equation is specified before any model is trained.
2. **A scaling coordinate.** Density, particle size, Knudsen number, wave nonlinearity, or scale separation can be swept toward and away from the trusted limit.
3. **Observable discarded information.** Cumulants, histories, non-equilibrium moments, and hierarchy residuals can be estimated from ensembles rather than discussed only qualitatively.
4. **A long-time test.** The results are designed to control kinetic behavior beyond the very short horizons on which a closure may look accurate accidentally.
5. **Controlled failure regimes.** One can deliberately leave the asymptotic assumptions by increasing density, strengthening correlations or coherent waves, reducing scale separation, or introducing steep kinetic structure.
6. **Error separation.** Numerical discretization, ensemble estimation, asymptotic-limit error, closure-model error, and learned-model error can be varied and reported separately.

3.3 From proof objects to a computational evaluation harness

For each canonical track, let U^ε denote the high-fidelity state at scale parameter ε, let $u^\varepsilon = \Pi U^\varepsilon$ be the retained state, and let $\mathcal{F}_0(u^\varepsilon)$ denote the declared classical reduced operator. The measurable closure defect is

$$\mathcal{D}^\varepsilon(t) = \partial_t u^\varepsilon(t) - \mathcal{F}_0(u^\varepsilon(t)). \tag{14}$$

The numerical program asks whether $\mathcal{D}^\varepsilon$ is predicted by a compact state built from projected cumulants, interaction-history summaries, non-equilibrium moments, or finite memory. Analytic, learned, and hybrid closures are then invoked through the same interface and scored with the same accuracy, stability, structure, reliability, and cost metrics.

The proof machinery is not itself assumed to be an efficient algorithm. Molecules, gardens, and cutting arguments first tell us what must be controlled. The computational question is whether low-order projections or learned compressions of those objects predict closure failure and improve the reduced solver. Negative answers are scientifically useful: they show that the proposed cumulant order, memory length, or representation is insufficient.

3.4 What transfers to TEMPEST—and what does not

The transferable contribution is the *experimental logic*: declare the hierarchy, identify the classical limit, expose the discarded correlations or moments, measure the exact closure defect, sweep a validity parameter, and test both asymptotic recovery and controlled breakdown. The common computational evaluation harness can carry that logic unchanged.

The physical equations and theorem assumptions do not transfer unchanged. TEMPEST applications may be magnetized, driven, bounded, multispecies, anisotropic, nonlocal, shock forming, or coupled across several physics packages. Their correlations need not be small, their memory may remain order one, and their natural reduced operator may be an effective dusty-particle model, a kinetic or moment model, a filtered turbulence model, a one-dimensional supernova carrier, radiation hydrodynamics, or another application-specific model rather than Boltzmann or the homogeneous wave kinetic equation.

The correct role of Hani's work in the bake-off

The canonical tracks validate whether the harness can detect, represent, and score closure in regimes where the reduction is mathematically understood. They do not claim that a hard-sphere or homogeneous-wave closure is a dusty-plasma or supernova closure. Each TEMPEST application must replace the equations, state, scaling parameters, constraints, observables, and coefficients while retaining the validated diagnostic and evaluation protocol. In the dusty-plasma profile, the discarded information includes plasma, wake, and field degrees of freedom. In the supernova profile, the discarded information includes nonradial turbulence, Reynolds stresses, shock corrugation, subgrid transfer, mixing, and history when a three-dimensional reference is projected into a one-dimensional carrier.

4 Machine learning for classical many-body systems

4.1 Scope: many-body, classical, and deliberately non-quantum

Many-body learning is sometimes treated as a single subject because all of its examples contain many interacting degrees of freedom. That grouping is too coarse for closure science. A method that identifies a radial pair force from particle tracks, a graph network that predicts the next particle state, a force-matched coarse potential, and a model that closes a kinetic moment hierarchy all use many-body data, but they approximate different mathematical objects and support different scientific claims.

This section is restricted to *classical* systems: interacting particles or agents, classical molecular and coarse-grained dynamics, granular and active matter, collective motion, point particles on manifolds, and laboratory dusty plasmas. It excludes wavefunction and density-matrix models, quantum Monte Carlo, quantum-state tomography, electronic-structure prediction, and machine-learned interatomic potentials whose target is a quantum Born–Oppenheimer energy surface. Classical molecular dynamics remains in scope when the learned object is a classical coarse free energy, stochastic reduced dynamics, or an interaction law inferred from classical trajectories.

A useful starting point is a heterogeneous second-order particle system

$$m_i \ddot{x}_i = F_i^{\mathrm{env}}(x_i, v_i, s_i; \mu) + \sum_{j \neq i} F_{ij}(x_i, x_j, v_i, v_j, s_i, s_j; \mu) - \Gamma_i(v_i, s_i; \mu) + \eta_i(t), \tag{15}$$

where s_i denotes particle type or latent attributes, μ denotes environmental parameters, and η_i represents unresolved forcing or thermal noise. Depending on the physics, F_{ij} may be reciprocal or nonreciprocal, conservative or dissipative, pairwise or effectively many-body, isotropic or anisotropic, instantaneous or history dependent. The state may live in Euclidean space, on a periodic domain, or on a manifold.

Even equation (15) is already a model reduction. Plasma particles may interact through an evolving wake or field; colloids through a solvent; active particles through a driven medium; coarse molecular beads through degrees of freedom that have been integrated out. The apparent force law can therefore be state dependent, stochastic, non-Markovian, and thermodynamic-state dependent. That observation is the first connection to closure: a learned effective force is often a closure for eliminated media or internal coordinates, not a microscopic fundamental law.

4.2 The learned object determines the scientific claim

The most important taxonomy is not neural network versus kernel method. It is the object being estimated.

Learned object	**Typical supervision**	**What success establishes**	**What it does not establish by itself**
Interaction kernel $\phi(r)$	Positions, velocities, or trajectory increments	A specified pairwise law is identifiable on sampled separations	Correct coarse statistics or validity outside the sampled occupancy measure
Force decomposition $F_{ij}, F_i^{\mathrm{env}}, \Gamma_i$	Particle tracks plus physical roles and descriptors	Interpretable effective forces under a declared decomposition	Uniqueness when components share inputs or compensating errors are possible
Latent interaction graph	Multivariate trajectories	Which components appear coupled under a graph model	A unique physical force law or causal mechanism
Vector field or time-step map	State-transition pairs	Predictive microscopic or mesoscopic dynamics	A closed reduced equation for moments or distributions
Coarse free energy or mean force	Fine-scale configurations and projected forces	Equilibrium coarse statistics under a stated mapping	Correct kinetics unless memory and fluctuating forces are also modeled
Kinetic or moment closure	Marginals, moments, cumulants, or closure defects	Evolution of a declared reduced state	Particle-level identifiability or faithful microscopic trajectories
Ensemble density or sampler	Equilibrium or stationary samples, possibly energies	A distribution over configurations or states	Dynamical fidelity, transport coefficients, or time correlations

Interaction discovery is not automatically closure

Suppose a model accurately recovers F_{ij} in equation (15). It has learned a law for advancing the retained particle state. A kinetic or fluid closure asks a different question:

$$\partial_t u = \mathcal{F}_0(u) + \mathcal{C}(u, \mathrm{history}, \mathrm{statistics}),$$

where $u = \Pi X$ retains only a marginal, moment, spectrum, or coarse observable. The map F_{ij} becomes a closure only after the eliminated variables, projection Π, and reduced operator $\mathcal{F}_0$ are specified. The same particle data can support both tasks, but the targets and validation tests are not interchangeable.

4.3 How many-body learning becomes closure: a summary strategy

A many-body learning method becomes a closure method when it is attached to a declared reduction. Let X be the higher-fidelity state, $u = \Pi X$ the retained state, and $\mathcal{F}_0$ the baseline reduced operator. The closure target is

$$\mathcal{C}^\star[X] = \Pi\mathcal{F}(X) - \mathcal{F}_0(\Pi X). \tag{16}$$

The learned model receives a declared information set $I[u]$ and approximates $\mathcal{C}^\star$, not an unspecified notion of missing physics:

$$\partial_t u = \mathcal{F}_0(u) + \mathcal{C}_\theta(I[u])\,. \tag{17}$$

Four-step closure strategy for a many-body method

1. **Declare the retained state.** Is the model advancing every particle, a one-particle distribution, a few moments, or fluid fields?
2. **Name the discarded information.** This may be a mediating field or bath, particle identities, connected correlations, higher moments, fast coordinates, or interaction history.
3. **Identify the learned object.** State whether the model estimates an effective pair force, graph message, mean force, memory kernel, collision operator, cumulant correction, stress, or heat flux.
4. **Validate at the reduced level.** Test the learned component directly where possible, then embed it in the reduced solver and measure stability, long-time statistics, structure, and recovery of the trusted limit.

The resulting scale ladder is

$$\begin{gathered}\text{particle tracks} \longrightarrow \text{effective interactions} \longrightarrow \text{particle simulator}\\ \longrightarrow \text{kinetic statistics} \longrightarrow \text{moment or fluid closure.}\end{gathered}$$

Errors can enter at every arrow. The evaluation harness should therefore preserve intermediate diagnostics rather than score only the final macroscopic output.

The same dusty-plasma experiment supports two distinct closure questions. If the retained state consists of dust-particle positions and velocities, then ions, electrons, fields, and wake dynamics are eliminated; a learned directed effective force is a *particle-level closure*. If the retained state is instead a dust distribution or a small set of collective moments, then particle identities, pair correlations, and non-equilibrium velocity structure are also eliminated; the required object is a *kinetic or moment closure*.

Retained state	Discarded information	Learned closure object	Decisive validation
Dust-particle state	Ions, electrons, plasma fields, and wakes	Directed pair force, environmental force, drag, or finite-memory state	Independent masses, charges, screening lengths, and particle rollouts
Dust distribution or moments	Particle identity, correlations, high moments, and particle histories	Correlation, collision, stress, transport, or memory correction	Distribution evolution, cumulants, transport, balance laws, and stable reduced rollouts

Thus, learning interactions and learning closure are not competing descriptions. Interaction learning can supply the first effective reduction, generate controlled particle ensembles, and expose the correlation and memory defects that must be closed at the next scale.

4.4 Structure-first nonparametric interaction-kernel inference

4.4.1 Why the apparent high dimension can be misleading

For N particles in $\mathbb{R}^d$, the full configuration lies in $\mathbb{R}^{Nd}$. Directly learning an unconstrained vector field on that space confronts both a large input dimension and a changing state dimension when N changes. In many classical systems, however, the governing law has repeated low-dimensional structure. A canonical first-order model is

$$\mathrm{d}x_i(t) = \frac{1}{N}\sum_{j=1}^{N} \phi(\|x_j(t) - x_i(t)\|)\,\big(x_j(t) - x_i(t)\big)\,\mathrm{d}t + \sigma\;\mathrm{d}B_i(t). \tag{18}$$

The unknown is the one-dimensional radial function $\phi(r)$, not a generic function on $\mathbb{R}^{Nd}$. Permutation symmetry, shared interactions, and radial dependence are therefore not cosmetic architectural preferences. They convert a high-dimensional regression problem into a structured inverse problem.

Lu et al. (2019) developed this nonparametric viewpoint for deterministic systems, and Lu et al. (2021a) extended it to multiple agent types and potentially non-symmetric cross-type kernels. The stochastic theory in Lu et al. (2022), one of the three anchor papers for this survey, treats multiple independent trajectories of equation (18). Because the drift is linear in ϕ, the likelihood objective becomes a quadratic functional, so a regularized maximum-likelihood estimate can be computed as a structured least-squares problem over a data-adaptive hypothesis space.

4.4.2 The occupancy measure is part of the answer

The natural error norm is weighted by the distribution of pairwise distances actually visited by the dynamics. Schematically,

$$\rho_T(A) = \frac{1}{T}\int_0^T \mathbb{E}\left[\frac{1}{N(N-1)}\sum_{i\neq j} \mathbf{1}_{\{\|x_j(t)-x_i(t)\|\in A\}}\right]\mathrm{d}t. \tag{19}$$

An estimate can be excellent in $L^2(\rho_T)$ while unconstrained on distances that never occur. This is not a technical nuisance: it precisely states where the experiment identifies the force law. A train/test split over trajectories does not test extrapolation in separation if both splits have nearly the same ρ_T. A serious benchmark should therefore report coverage in separation, relative speed, particle type, height, field strength, or whichever variables parameterize the candidate interaction.

The associated coercivity condition says, roughly, that distinct candidate kernels produce detectably distinct collective drifts over the observed ensemble. For a hypothesis space $\mathcal{H}$, it takes the schematic form

$$\mathbb{E}\left[\frac{1}{N}\sum_{i=1}^{N}\left|\frac{1}{N}\sum_{j=1}^{N}\varphi(r_{ij})(x_j - x_i)\right|^2\right] \geq c_{\mathcal{H}}\|\varphi(\cdot)\cdot\|^2_{L^2(\rho_T)}. \tag{20}$$

The constant $c_{\mathcal{H}} > 0$ measures identifiability and conditioning. Small values mean that particle-wise contributions cancel or that the observed configurations do not sufficiently excite the candidate modes. This gives a more useful experimental-design question than asking only how many frames are available: do the trajectories explore configurations that distinguish the interaction laws of interest?

Under regularity and coercivity assumptions, the estimator in Lu et al. (2022) attains the one-dimensional nonparametric rate

$$\left(\frac{\log M}{M}\right)^{s/(2s+1)} \tag{21}$$

for M independent trajectories and smoothness s, independent of the ambient configuration dimension Nd. The dimension independence is purchased by the correctness of the structural ansatz. If the true dynamics contain angular dependence, three-body terms, hidden types, or memory that the ansatz omits, the theorem does not make those discrepancies disappear.

4.4.3 Time resolution is an information budget

The stochastic paper also isolates an error that is easy to bury inside a generic data-noise category. When only discrete-time positions are available and an Euler–Maruyama likelihood approximation is used, the estimator incurs a bias of order

$$\sigma\, O(\Delta t^{1/2}). \tag{22}$$

Once this term dominates sampling and approximation errors, adding more trajectories no longer improves the observed learning curve. The numerical experiments show this flattening explicitly. For the bake-off, temporal resolution must therefore be reported alongside trajectory count, and synthetic data should distinguish simulator time step from observation gap.

The paper's numerical results also connect kernel error in $L^2(\rho_T)$ to trajectory discrepancy and show good collective predictions beyond the training interval in its tested stochastic opinion and Lennard–Jones systems. This is valuable evidence, but it does not eliminate the need for an embedded long-time test: stable collective behavior can be more forgiving than pointwise particle paths, and both should be measured.

4.5 Geometry is part of the physics

The second anchor paper, Maggioni et al. (2021), considers particles constrained to a connected, smooth, geodesically complete Riemannian manifold $\mathcal{M}$. The Euclidean distance and displacement in equation (18) are replaced by geodesic distance and the tangent direction of a shortest geodesic:

$$\dot{x}_i = \frac{1}{N}\sum_{j=1}^{N} \phi(d_{\mathcal{M}}(x_i, x_j))\, w(x_i, x_j), \qquad \dot{x}_i \in T_{x_i}\mathcal{M}. \tag{23}$$

Here $w(x_i, x_j)$ is a tangent vector at x_i pointing locally toward x_j. The estimator, coercivity condition, occupancy measure, and trajectory error are all defined using the manifold geometry.

The examples include opinion and predator–swarm dynamics on the two-sphere and the Poincare disk. The estimator again achieves a convergence rate independent of Nd, and a kernel learned at one particle number can be reused at another because the object being learned is shared rather than tied to a fixed-dimensional state vector. This is a concrete form of structural transfer.

The lesson extends well beyond curved surfaces. Periodic coordinates, constrained rigid bodies, magnetic field-line geometry, conservation manifolds, and quotienting out translations or center-of-mass motion all change the admissible state and the metric used to compare trajectories. Projecting a generic neural output back onto a constraint surface after each step is not always equivalent to learning a tangent vector field. Geometry must be specified in the model interface, the loss, and the evaluation metric.

What the kernel-inference papers contribute

The Lu–Maggioni program provides more than an estimator. It supplies a template for mathematically auditable many-body learning:

1. identify a shared low-dimensional interaction object;
2. state the symmetry and geometry that make it shared;
3. define the dynamics-adapted occupancy measure;
4. test a coercivity or excitation condition;
5. separate sampling, approximation, observation-gap, and rollout errors; and
6. state explicitly where the assumed interaction family can fail.

4.6 Physics-tailored neural force discovery in dusty plasma

4.6.1 Why this is a stronger experimental test

The third anchor paper moves from controlled simulated systems to laboratory trajectories. Yu et al. (2025) study micron-scale charged dust particles levitated in a weakly ionized argon plasma. The background ions and electrons mediate effective particle interactions. Ion flow creates wakes below the particles, so the horizontal interaction can depend on vertical ordering, can be attractive in some configurations, and need not satisfy reciprocity $F_{ij} = -F_{ji}$. Particle sizes are heterogeneous, the confining environment is spatially varying, and only noisy trajectory observations are available.

These features make dusty plasma unusually relevant to TEMPEST. It is classical and particle resolved, yet its effective interactions already encode eliminated plasma degrees of freedom. It supports controlled laboratory validation, non-equilibrium and nonreciprocal physics, heterogeneous particles, variable particle counts, and a natural route from individual interactions to collective statistics.

4.6.2 The decomposition is the main inductive bias

The measured horizontal dynamics are organized as

$$\ddot{\boldsymbol{\rho}}_i = \sum_{j \neq i} f_{ij}\, \widehat{\boldsymbol{\rho}}_{ij} + \boldsymbol{f}_i^{\text{env}} - \gamma_i \dot{\boldsymbol{\rho}}_i, \tag{24}$$

where $\boldsymbol{\rho}_i = (x_i, y_i)$, while the learned pair force may depend on the horizontal separation ρ_{ij}, the two vertical positions z_i, z_j, and particle descriptors s_i, s_j. Three networks are trained jointly but assigned different physical roles:

$$f_{ij} = g_{\text{int}}(\rho_{ij}, z_i, z_j, s_i, s_j), \tag{25}$$

$$\boldsymbol{f}_i^{\text{env}} = g_{\text{env}}(x_i, y_i, z_i, s_i), \tag{26}$$

$$\gamma_i = g_\gamma(s_i). \tag{27}$$

The descriptor $s_i = \langle z_i \rangle_t$ acts as a proxy for particle size and mass because heavier particles settle lower in the sheath. The pair network preserves horizontal translation symmetry while allowing broken vertical translation symmetry; its directed inputs allow nonreciprocal interactions. The environmental and damping networks have different input sets because otherwise the three terms in equation (24) would be free to compensate for one another and the decomposition would be poorly identifiable.

This is a particularly clear example of a physics-tailored neural architecture. The network is flexible where the force law is unknown, but the decomposition, symmetries, direction of the horizontal pair force, drag form, and role-specific inputs are declared in advance. The model also accepts varying particle number because pair contributions are shared and aggregated.

4.6.3 Weak-form learning and independent validation

Directly differentiating noisy position tracks twice is ill conditioned. The dusty-plasma model instead uses a weak-form loss: trajectories are integrated against compact test functions, and integration by parts moves derivatives away from the data. This is the same numerical principle that motivates weak-form system identification such as WSINDy (Messenger and Bortz, 2021), although the hypothesis class and loss construction differ. The weak form reduces derivative noise without manufacturing information that is absent from the observations.

The model attains held-out acceleration coefficients of determination above 0.99 across experiments. Crucially, the authors do not treat that score as proof that every force component is correct. The total acceleration is a sum of three learned contributions, so compensating errors could produce an excellent aggregate fit. They therefore estimate particle mass in two independent ways, using learned interaction forces and learned drag, and compare the inferred values with particle sizes. For approximately coplanar pairs they also fit a screened Coulomb form to infer masses, charges, and screening lengths.

Those secondary tests turn a predictor into a scientific instrument. The analysis supports nonreciprocal and attractive interactions, finds that the effective screening length changes with the average particle size rather than depending only on nominal plasma conditions, and finds a charge–mass exponent that varies with gas pressure instead of remaining at the simplest 1/3 scaling. The lesson is general:

$$\begin{aligned}&\text{fit of total acceleration} \;<\; \text{component validation}\\ &<\; \text{independent physical recovery} \;<\; \text{intervention or new-regime prediction.}\end{aligned}$$

4.6.4 Limits of the result

The model learns horizontal reduced forces in the experimental regime occupied by systems of roughly ten to twenty tracked particles. Pair evaluation scales quadratically with particle count. The assumed radial horizontal direction, role decomposition, descriptor sufficiency, and instantaneous pairwise form remain hypotheses. A plasma-mediated interaction that depends on collective wake state, delayed response, or three-body configuration could be absorbed into an effective pair network over the training distribution and fail under a new particle density or forcing history.

The appropriate next step is therefore not simply a larger network. It is a hierarchy of falsification tests: vary particle count and density, hold out vertical-ordering regimes, compare pairwise and message-passing models, add finite memory, test force reciprocity only where theory predicts it, and ask whether inferred parameters remain consistent across independent experimental protocols.

4.7 Graph and message-passing models

4.7.1 Particles as nodes and interactions as messages

Graph neural networks replace a fixed-size state vector by nodes, edges, and permutation-equivariant aggregation. A generic message-passing layer has the form

$$m_{ij}^{(\ell)} = \psi_e^{(\ell)}\big(h_i^{(\ell)}, h_j^{(\ell)}, e_{ij}, x_i - x_j\big), \tag{28}$$

$$\bar{m}_i^{(\ell)} = \sum_{j \in \mathcal{N}(i)} m_{ij}^{(\ell)}, \tag{29}$$

$$h_i^{(\ell+1)} = \psi_v^{(\ell)}\big(h_i^{(\ell)}, \bar{m}_i^{(\ell)}\big). \tag{30}$$

Shared edge functions express repeated interactions; summation gives permutation equivariance and variable particle number. Neighbor lists, cutoff graphs, or learned edges determine computational cost and locality.

Neural relational inference places a latent graph between observed trajectories and a graph-based decoder (Kipf et al., 2018). It is useful when the interaction topology or edge types are unknown, but an inferred edge label should not automatically be read as a physical bond or causal mechanism. Latent relations can absorb unmodeled common forcing, observation artifacts, and multi-step correlations.

Graph Network-based Simulators train on one-step particle accelerations and roll the predictions forward (Sanchez-Gonzalez et al., 2020). The reported framework handles fluids, deformable materials, and rigid objects, and generalizes in its tests to longer rollouts and more particles than used in training. Training-time noise corruption reduces the train–rollout distribution mismatch. This is an important numerical point: a small one-step error is repeatedly fed back as a new input, so long-rollout stability is a property of the learned model plus integrator, not of the regression score alone.

4.7.2 Equivariance beyond permutation symmetry

Standard message passing is permutation equivariant but not automatically rotation or translation equivariant in geometric outputs. $E(n)$-equivariant graph networks construct coordinate and feature updates that commute with Euclidean translations, rotations, and reflections (Satorras et al., 2021). For force and dynamics problems, the desired group must be chosen rather than assumed. Reflections may or may not be a symmetry; magnetic fields, chirality, walls, gravity, or driven media can select orientations. Dusty plasma, for example, has horizontal translation structure but a distinguished vertical direction and nonreciprocal directed interactions.

Equivariance restricts the hypothesis space and can improve data efficiency, but it does not guarantee conservation, stability, correct asymptotics, or correct interactions. A model can be perfectly rotation equivariant and learn the wrong radial function. Symmetry is one structural test, not a substitute for physics-specific validation.

4.8 Sparse and symbolic discovery

Sparse identification of nonlinear dynamics assumes that the governing vector field is a sparse combination of candidate functions (Brunton et al., 2016). For many-body systems, a naive library over all particle coordinates grows rapidly and obscures shared structure. A better construction builds the library from invariant or equivariant quantities such as distances, relative velocities, types, local densities, and field-aligned components.

Weak-form SINDy replaces pointwise derivative estimates by integrals against test functions and can remain effective under substantial measurement noise (Messenger and Bortz, 2021). At the particle-to-continuum level, Messenger and Bortz (2022) combine mean-field theory with weak sparse identification to learn candidate mean-field equations directly from particle data. Their method targets a population-level equation rather than each microscopic force and demonstrates $O(N^{-1/2})$ convergence of a least-squares component with particle number under stated assumptions. This is especially relevant to closure because it crosses a scale rather than only emulating particle dynamics.

A complementary strategy first fits an expressive graph model and then applies symbolic regression to its learned messages or energy (Cranmer et al., 2020b). The graph provides a structured high-capacity intermediate representation; symbolic regression seeks a compact analytic law. This can improve interpretability and extrapolation when the distilled expression is correct. It also introduces a second selection problem: the symbolic search can recover an attractive expression that fits the network rather than the physical data. The final expression must be re-evaluated directly on held-out trajectories, invariants, parameter sweeps, and rollouts.

Sparse and symbolic methods are strongest when the relevant coordinates and candidate operations are approximately known. Failure to recover a compact formula can mean insufficient data, a poor library, hidden variables, or a genuinely nonlocal or history-dependent closure. Those possibilities should not be collapsed into a claim that the dynamics are not interpretable.

4.9 Structure-preserving learned dynamics

For an isolated conservative system in canonical coordinates $z = (q, p)$, a Hamiltonian model writes

$$\dot{z} = J\nabla_z H_\theta(z), \qquad J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}. \tag{31}$$

Hamiltonian neural networks learn the scalar H_θ and obtain the vector field through this antisymmetric structure (Greydanus et al., 2019). Lagrangian neural networks instead learn $L_\theta(q, \dot{q})$ and use the Euler–Lagrange equations, avoiding a requirement for observed canonical momenta (Cranmer et al., 2020a). Graph versions decompose energy into shared particle and interaction contributions.

These are hard structural priors only relative to their assumptions. They are appropriate for closed conservative mechanics, but not for a driven dusty plasma with drag, environmental forcing, and energy supplied through nonreciprocal wake interactions. Applying a conservative architecture there would force nonconservative effects into state dependence or model error. More suitable structures include conservative-plus-dissipative splits, port-Hamiltonian forms, metriplectic or GENERIC constructions, constrained tangent dynamics, or explicit forcing and bath variables.

The numerical integrator remains part of the method. A continuous Hamiltonian vector field integrated by a generic solver need not preserve a discrete symplectic form exactly; conversely, a learned symplectic time-step map may preserve discrete geometry without identifying the underlying force. Benchmark reports should name both the learned object and the stepping scheme.

4.10 Classical coarse-graining, memory, and closure

4.10.1 Coarse free energies

Let X be a fine classical state and $z = \xi(X)$ a coarse coordinate. Bottom-up coarse-graining seeks an effective free energy $U_{\mathrm{cg}}(z)$ or mean force whose equilibrium distribution matches the projected fine system. Force matching trains on projected fine-scale forces. CGnets use neural potentials with translation and rotation structure, conservative forces, and prior energies that discourage unphysical configurations outside the data support (Wang et al., 2019).

This is genuine closure: eliminated coordinates contribute a potential of mean force that is generally many-body even if the microscopic forces were pairwise. Yet equilibrium representability and dynamical fidelity are different. A coarse potential can reproduce the equilibrium marginal while producing incorrect rates, time correlations, and transport because eliminated variables also generate friction, memory, and fluctuating forces.

4.10.2 Memory is expected, not exceptional

For a projection $z = \Pi X$, Mori–Zwanzig theory organizes exact reduced dynamics into a Markov term, a memory convolution, and an orthogonal or fluctuating contribution. Data-driven model reduction can therefore fit finite-memory, autoregressive, auxiliary-variable, or generalized Langevin representations rather than pretending the coarse process is Markovian. Lin and Lu (2021) connect Wiener projection, Koopman operators, Mori–Zwanzig reduction, and NARMAX models for deterministic chaotic and randomly forced systems.

In a many-body benchmark, the Markov-versus-memory comparison should be explicit:

$$\dot{z}(t) = F_\theta(z(t)), \tag{32}$$

$$\dot{z}(t) = F_0(z(t)) + \int_0^t K_\theta(z(t-s), s)\,\mathrm{d}s + \eta_\theta(t), \tag{33}$$

$$\dot{z}(t) = F_0(z(t)) + C_\theta(z(t), h(t)), \qquad \dot{h} = G_\theta(z, h). \tag{34}$$

The auxiliary state h is often the most practical finite-dimensional realization of memory. It should be justified by improved long-time statistics or prediction at fixed information and compute budgets, not merely by a lower training loss.

4.11 Learning ensembles rather than trajectories

Some scientific questions concern configuration distributions, rare states, or equilibrium expectations rather than time-ordered paths. Normalizing flows learn an invertible map from a simple base density to a many-body distribution. Equivariant flows build symmetries of the target density into this map and can improve sampling and generalization in symmetric classical particle systems (Köhler et al., 2020).

An ensemble model can supply equilibrium samples, likelihoods, free-energy estimates, or initial conditions for rare-event studies. It is not a dynamics model unless additional temporal structure is learned. Two simulators can have the same stationary distribution and different kinetics; two dynamics models can produce similar short trajectories and different invariant measures. The harness should therefore keep configuration-distribution metrics separate from trajectory and transport metrics.

4.12 A validation ladder for classical many-body learning

Many published demonstrations stop at one-step error, acceleration R^2, or visual plausibility. Closure applications require a stricter ladder.

1. **Observation fit.** Held-out position-increment, velocity, acceleration, or weak-residual error.
2. **Component recovery.** Error in pair forces, environment forces, drag, graph edges, or energy when ground truth exists.
3. **Independent physical validation.** Recovery of masses, charges, screening lengths, friction, equation coefficients, or controlled-response curves not used directly as labels.
4. **Short rollout.** Trajectory error with the model embedded in the declared numerical integrator.
5. **Long-time behavior.** Stability, collisions, boundedness, invariant measure, correlations, transport, and collective modes.
6. **Structural tests.** Symmetry or equivariance, constraints, conservation or balance laws, dissipation, reciprocity where applicable, and dimensional consistency.
7. **Scale transfer.** Held-out particle number, density, geometry, types, sampling interval, or domain size.

8. **Regime transfer.** Held-out forcing, field strength, pressure, temperature, collisionality, wake strength, or scale-separation parameter.

9. **Closure test.** Accuracy of the induced reduced operator, cumulants, constitutive fluxes, and macroscopic observables after projection.

The uncertainty record should separate at least

$$\begin{aligned}&\text{tracking error} + \text{derivative or weak-window error} + \text{finite-ensemble error}\\&\quad + \text{time-discretization error} + \text{model-form error} + \text{rollout error}.\end{aligned} \tag{35}$$

For experiments, particle identity mistakes and missing tracks are additional structured errors. For stochastic systems, pathwise divergence is not by itself failure; distributional and statistical metrics must accompany particle-wise errors.

4.13 Common failure modes

Pairwise misspecification
An effective pair model can absorb three-body, density-dependent, field-mediated, or collective effects within one training regime and then fail when density or geometry changes. Compare pairwise kernels with local-density corrections and multi-message graph models.

Hidden heterogeneity
Treating nonidentical particles as exchangeable can broaden a learned force law and create false stochasticity. Type labels or identifiable descriptors may be required, but a descriptor correlated with location can also confound particle and environmental effects.

Unobserved state and memory
If wakes, solvent modes, internal orientations, or fields evolve on comparable time scales, an instantaneous function of visible positions and velocities is not closed. Lag tests and auxiliary-state models should be predeclared.

Cancellation and non-identifiability
Accurate total acceleration does not guarantee accurate force components. Role-specific inputs, interventions, excitation design, coercivity diagnostics, and independent parameter recovery are remedies.

Symmetry mismatch
Hard-coding reciprocity, isotropy, reflection symmetry, or energy conservation where the driven environment breaks it produces systematic bias. Omitting a true symmetry wastes data and permits inconsistent predictions.

Derivative amplification
Finite differences amplify tracking noise, especially for acceleration. Weak forms reduce this amplification but introduce window and test-function choices that must be reported and swept.

Quadratic cost and long-range interactions
All-pairs evaluation costs $O(N^2)$. Cutoffs, neighbor lists, multipole summaries, hierarchical graphs, or learned field representations can reduce cost, but each changes the information available to the model.

Rollout distribution shift

Training states come from the reference dynamics; rollout states increasingly come from the learned dynamics. Noise injection, multi-step losses, invariant penalties, stable integration, and recovery mechanisms address different parts of this problem.

4.14 A concrete many-body ladder for the TEMPEST bake-off

The dusty-plasma profile can turn this literature into a controlled experiment rather than a collection of architectures. The common computational evaluation harness supplies fixed data products, splits, integration, scoring, and cost accounting. Each candidate implements one declared interface and receives only its predeclared information set.

Rung	Candidate	Learned object	Decisive test
0	Screened analytic pair law plus drag and confinement	Calibrated coefficients	Trusted-regime recovery and transparent failure boundary
1	Nonparametric structured kernel estimator	Directed type- and state-dependent pair kernels	Identifiability, occupancy coverage, and kernel recovery on synthetic controls
2	Yu-style physics-tailored networks	Pair force, environment force, and drag	Independent mass/charge validation and held-out experimental regimes
3	Equivariant or field-aware graph model	Multi-particle vector field or acceleration	Benefit of higher-order context at matched data and compute
4	Finite-memory graph or auxiliary-state model	Non-Markovian particle dynamics	Improvement under forcing-history and wake-memory tests
5	Mean-field or kinetic equation discovery	Population-level operator	Recovery of density evolution, pair correlations, and transport across N
6	Learned cumulant or moment closure	Correction to an explicit reduced operator	A priori closure-defect accuracy and a posteriori stable reduced simulation

The primary comparison should not ask which rung wins a single scalar metric. It should ask how much structure and information are required for each scientific objective. Rungs 1–3 test interaction discovery and particle simulation. Rungs 5–6 test scale transition and closure. Rung 4 tests whether visible particle state is Markovian. Rung 0 anchors interpretability and sample efficiency.

Recommended split axes are particle identity, particle count, pressure, magnetic field, vertical separation and ordering, pair-distance occupancy, initial condition, trajectory segment, and forcing history. Synthetic data with a known force decomposition should precede experimental fitting so that identifiability and component recovery can be measured. Experimental conclusions should require at least one validation quantity not used in training.

How this serves the broader TEMPEST program

The many-body survey supplies a reusable front end for any particle-resolved TEMPEST application:

$$\boxed{\begin{array}{c}\text{structured interaction discovery} \;\to\; \text{particle or kinetic model} \\ \to\; \text{measured closure defect} \;\to\; \text{structure-preserving reduced closure.}\end{array}}$$

The application physics determines the state, symmetries, interactions, reduced operator, and validation quantities. The computational/evaluative harness determines the interfaces, information budgets, matched comparisons, uncertainty decomposition, and reproducibility rules. Dusty plasma is therefore both a science application and a test of whether particle-level ML can be translated honestly into closure.

5 Neural networks as numerical approximation families

5.1 The finite-element analogy

A traditional numerical method begins by choosing an approximation family. A finite-element method writes

$$u_h(x,t) = \sum_{j=1}^{N} c_j \phi_j(x,t), \tag{36}$$

where the basis functions ϕ_j are prescribed and the coefficients c_j are determined from a discrete variational or algebraic problem. A spectral method uses Fourier modes or orthogonal polynomials. A radial-basis method uses translated kernels.

A neural approximation instead chooses a nonlinear parameterized family

$$u_\theta(x,t) = W_L\sigma\big(W_{L-1}\sigma(\cdots\sigma(W_1[x,t]+b_1)\cdots)+b_{L-1}\big)+b_L. \tag{37}$$

The parameters θ consist of weights and biases. The neural network is therefore not mysterious at this level: it is a large adaptive trial family built from compositions of affine maps and nonlinearities.

The important novelty is not merely that this family can approximate a function u. The same technology can parameterize an entire solver map, an unknown constitutive law, a vector field, an invariant coordinate system, or a transition kernel.

Approximation technology	**Typical representation**
Finite differences	Values on a grid and local difference stencils
Finite elements	Piecewise-polynomial trial and test spaces
Spectral methods	Fourier, Chebyshev, or other global basis expansions
Radial basis methods	Sums of translated kernels
Neural methods	Compositions of affine maps and nonlinear activations, possibly with convolution, attention, integral kernels, recurrence, or geometric constraints

The choice of learned object determines the training problem and the deployment model.

5.2 Approximating one function

A coordinate network may represent

$$u_\theta : (x,t) \longmapsto u_\theta(x,t).$$

This is the basic PINN viewpoint. The network is a trial function for one solution. The parameters are selected so that this trial function fits observations, initial and boundary conditions, and a differential-equation residual.

5.3 Approximating a map between functions

A neural operator represents

$$\mathcal{G}_\theta : a(\cdot) \longmapsto u(\cdot).$$

The input may be an initial condition, forcing, coefficient field, boundary function, geometry encoding, or a collection of parameters. The output is a full function. The goal is not one solution but a reusable approximation to a whole solution operator.

5.4 Approximating a missing term

A learned closure represents

$$\mathcal{C}_\theta : \text{resolved state and features} \longmapsto \text{unresolved flux, stress, or moment.}$$

The known equations remain in place. A conventional numerical method solves them after the network supplies the missing relation.

5.5 Approximating a vector field or flow map

A neural ODE learns

$$\dot{z} = F_\theta(z),$$

and then an ODE solver integrates the learned vector field. A learned flow map instead predicts

$$z^{n+1} = \Phi_\theta(z^n),$$

thereby replacing one time step or a finite-time evolution map directly.

These are not interchangeable. Learning F asks for an infinitesimal generator; learning $\Phi_{\Delta t}$ asks for a finite-time map. Their stability, data requirements, and dependence on the chosen time increment differ.

5.6 Approximating geometry or latent state

A network can learn a coordinate transformation

$$z \longmapsto (z_{\text{slow}}, z_{\text{fast}}),$$

an invariant manifold

$$z_{\text{fast}} = h_\theta(z_{\text{slow}}),$$

or an encoder from observation history to latent state

$$z_t = E_\theta(o_0, a_0, \ldots, o_t).$$

Here the network is not merely predicting the next physical field. It is constructing the state variables in which prediction or control becomes feasible.

6 Classical physics-informed neural networks

6.1 The narrow, original meaning

For a PDE

$$\mathcal{N}_\lambda[u] = f \quad \text{in } \Omega, \qquad \mathcal{B}[u] = g \quad \text{on } \partial\Omega, \tag{38}$$

a classical PINN chooses a coordinate network $u_\theta(x,t)$ and minimizes a loss of the form

$$\begin{aligned} \mathcal{L}(\theta) = \lambda_{\text{PDE}} \frac{1}{N_r} \sum_{j=1}^{N_r} \left| \mathcal{N}_\lambda[u_\theta](x_j, t_j) - f(x_j, t_j) \right|^2 \\ + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}} + \lambda_{\text{data}} \mathcal{L}_{\text{data}}. \end{aligned} \tag{39}$$

Automatic differentiation evaluates derivatives of u_θ. The network parameters are optimized so that one neural trial function approximately satisfies the equation, constraints, and data (Raissi et al., 2019).

The right mental model for a classical PINN

A classical PINN replaces a conventional discrete PDE solve by a nonlinear optimization problem over a neural trial space. It ordinarily learns *one solution* for one problem instance.

The same residual-based idea can be used for ODEs, differential-algebraic equations, kinetic equations, and some integro-differential or stochastic problems. Thus the phrase "PINN" need not literally require a partial differential equation. What it requires in its classical form is a known governing equation whose residual can be evaluated.

6.2 Forward and inverse problems

For a forward problem, the equation and coefficients are known and u is unknown. For an inverse problem, one may learn both u_θ and unknown parameters λ, for example

$$\partial_t u + \lambda u \partial_x u - \nu \partial_{xx} u = 0.$$

Sparse observations can inform λ and ν while the residual regularizes the inferred state.

This is one of the most natural uses of PINNs: the unknown state and unknown physical parameters are fitted jointly from incomplete observations under a known equation form.

6.3 Where the physics enters

In a vanilla PINN, the physics is primarily a *soft penalty*:

$$\mathcal{N}[u_\theta] \approx 0.$$

The optimizer may trade residual accuracy against initial conditions, boundary conditions, and data fit. A small sampled residual does not automatically imply

$$\begin{gathered}\text{exact conservation}, \qquad \text{positivity}, \qquad \text{hyperbolicity}, \\ \text{entropy stability}, \qquad \text{correct long-time dynamics}.\end{gathered}$$

Many refined PINN methods add hard boundary parameterizations, weak or variational forms, domain decomposition, adaptive sampling, conservative formulations, causal training, and special architectures. Consequently, one should not reduce all PINNs to the simplest squared-residual implementation. Nevertheless, the defining learned object remains a solution function, and the governing equation is normally used directly in training.

6.4 Why PINNs can be difficult

The PDE residual, boundary residual, initial residual, and data residual can operate on different scales and generate conflicting gradients. High frequencies, sharp fronts, stiffness, chaotic dynamics, and long time intervals can make the optimization poorly conditioned.

There is also a distinction between an approximation error and an optimization error. The network family may be expressive enough to represent a good solution, yet gradient-based training may not find it. Classical numerical analysis often studies consistency and stability of a specified discretization. PINNs add a large nonconvex optimization layer between the approximation family and the computed answer.

7 Neural operators

7.1 From one solution to a family of solutions

Suppose a denotes problem data:

$$a = \{\text{coefficient field, initial condition, forcing, boundary data, geometry, parameters}\}.$$

A PDE or another process defines a map

$$\mathcal{G}^\dagger : a(\cdot) \longmapsto u(\cdot). \tag{40}$$

A neural operator learns

$$\mathcal{G}_\theta \approx \mathcal{G}^\dagger$$

from many input-output pairs

$$\{a_j, u_j\}_{j=1}^N, \qquad u_j = \mathcal{G}^\dagger(a_j).$$

DeepONet, Fourier neural operators, graph neural operators, kernel neural operators, wavelet constructions, and operator transformers are architectural families for representing maps between functions (Lu et al., 2021b; Li et al., 2021a; Kovachki et al., 2023).

The right mental model for a neural operator

A neural operator pays a potentially large offline training cost to build an amortized solver for an entire distribution of problem instances. Once trained, a new input function can be mapped to an output function very quickly.

A standard supervised objective is

$$\min_\theta \frac{1}{N} \sum_{j=1}^{N} \|\mathcal{G}_\theta(a_j) - u_j\|_{\mathcal{U}}^2 . \tag{41}$$

The network is shared across many instances, unlike a classical PINN that is normally reoptimized for one instance.

7.2 A standard neural operator does not require knowing a PDE

The operator-learning problem is mathematically

$$\text{input function} \longmapsto \text{output function}.$$

Paired data are sufficient. The outputs could come from a PDE solver, laboratory measurements, molecular dynamics, a different simulator, or any reproducible function-to-function transformation. A PDE may have generated the data, but the training algorithm need not evaluate or even know it.

This yields an important distinction:

$$\begin{aligned} \text{PDE as training constraint:} \quad & \mathcal{N}[\mathcal{G}_\theta(a)] \text{ is evaluated;} \\ \text{PDE as data source:} \quad & \text{only pairs } (a, u) \text{ are used;} \\ \text{PDE as architectural inspiration:} \quad & \begin{array}{c} \text{conservation, locality, or multiscale structure} \\ \text{shapes the model.} \end{array} \end{aligned}$$

7.3 Fourier neural operators and discretization

An FNO alternates pointwise nonlinearities with global Fourier-domain integral operators. In simplified form,

$$v_{\ell+1}(x) = \sigma\Big(W_\ell v_\ell(x) + \mathcal{F}^{-1}\big(R_\ell(k)\mathcal{F}v_\ell(k)\big)(x)\Big) . \tag{42}$$

The operator viewpoint is intended to separate the learned map from one fixed grid, although actual performance still depends on sampling, truncation, architecture, and the input distribution.

The attractive computational economics are

$$\text{expensive data generation and training} \quad \longrightarrow \quad \text{very cheap repeated inference.}$$

This is particularly useful for uncertainty quantification, inverse design, optimization, data assimilation, control, and parameter sweeps requiring many related solves.

7.4 PINO: the bridge between PINNs and neural operators

A physics-informed neural operator combines operator regression with equation constraints:

$$\mathcal{L}_{\text{PINO}} = \mathcal{L}_{\text{operator data}} + \lambda_{\text{PDE}} \mathbb{E}_{a\sim\mu} \left[\|\mathcal{N}_a[\mathcal{G}_\theta(a)]\|^2 \right]. \tag{43}$$

It learns over many problem instances, but uses the PDE residual as supervision or regularization (Li et al., 2021b).

Thus:

	Learned object	**Training scope**
PINN	One solution u_θ	One instance, usually equation-constrained optimization
Neural operator	Operator $\mathcal{G}_\theta$	Many input-output instances, often supervised
PINO	Operator $\mathcal{G}_\theta$	Many instances with data and/or PDE constraints

7.5 Neural operators are not automatically physical

A neural operator can have low one-step or fieldwise error while violating conservation, positivity, causality, symmetry, or stability. Recursive rollouts may amplify small errors. The operator can also fail under distribution shift: a new coefficient field, initial condition, forcing, or geometry may lie outside the training measure.

The phrase "operator learning" therefore identifies the learned object. It does not by itself certify physical admissibility or long-time reliability.

8 Soft physics and hard physics

The contrast between residual PINNs and much structure-preserving scientific machine learning can be summarized as follows.

8.1 Soft physics

A loss term says

"Please make this violation small on sampled training states."

For example, one might penalize nonreal characteristic speeds by

$$\mathcal{L}_{\mathrm{hyp}} = \sum_q \left|\operatorname{Im} \lambda_q(A_\theta)\right|^2. \tag{44}$$

The optimizer can trade this penalty against other objectives, and the condition is checked only where training samples or collocation points are supplied.

8.2 Hard physics

An architectural parameterization says

"Every model produced by this architecture belongs to an admissible class."

Examples include

$$\begin{aligned}
A_\theta(m) &\text{ is hyperbolic by construction,}\\
(D\Phi_\theta)^T J (D\Phi_\theta) &= J \text{ by composition of symplectic layers,}\\
\mathcal{M}_\theta &\text{ is an attracting invariant manifold by design,}\\
\sum_i (u_i^{n+1} - u_i^n) &= 0 \text{ through a conservative projection.}
\end{aligned}$$

Hard structure does not guarantee predictive accuracy. A perfectly hyperbolic closure can be quantitatively wrong; a perfectly symplectic map can approximate the wrong dynamics. What it guarantees is that the learned model will not fail in that particular prohibited way.

The Christlieb–Tang methodological theme

Retain trusted mathematical structure, isolate the unknown or expensive component, parameterize only an admissible class for that component, and learn the remaining degrees of freedom from data.

9 Andrew Christlieb's program: learned kinetic closures inside admissible PDE models

Christlieb's work is the most direct continuation of the classical closure story. The network ordinarily does not represent the full kinetic solution and does not replace the entire reduced PDE solver. It supplies a carefully chosen missing relation in a moment system.

9.1 The kinetic moment problem

In one spatial and velocity dimension, let

$$m_k(x,t) = \int_{\mathbb{R}} v^k f(x,v,t)\,\mathrm{d}v. \tag{45}$$

Taking moments of a transport or kinetic equation produces a hierarchy resembling

$$\partial_t m_k + \partial_x m_{k+1} = S_k, \qquad k = 0, 1, \ldots. \tag{46}$$

If one retains $m_0, \ldots, m_N$, the last equation contains the unknown m_{N+1}. A closure is needed.

A straightforward learned closure would predict

$$m_{N+1} = \mathcal{C}_\theta(m_0, \ldots, m_N).$$

Christlieb and collaborators instead developed a gradient-based relation of the form

$$\partial_x m_{N+1} \approx \sum_{j=0}^{N} a_{j,\theta}(m_0, \ldots, m_N)\, \partial_x m_j. \tag{47}$$

The network predicts the coefficients $a_{j,\theta}$, and the closed moment PDE is advanced by a numerical method.

What the neural network is approximating

In this program the network approximates neither $f(x, v, t)$ nor the full input-to-solution map. It approximates the missing high-moment flux contribution needed by the last retained moment equation.

This is a genuine constitutive closure:

$$\boxed{\text{known moment equations} + \text{learned top-order closure} + \text{conventional hyperbolic solver.}}$$

9.2 Radiative-transfer closure I: learn the gradient relation

The first paper in the radiative-transfer series proposes learning the gradient of the unclosed moment rather than learning the moment itself (Huang et al., 2021a). This choice is motivated by the exact free-streaming structure and by the role the missing derivative plays in the retained PDE.

The conceptual move is important. A generic regression problem would ask the network to approximate an available target. A closure-oriented design asks a more structural question:

> Which unresolved quantity appears in the reduced equation, and what is the smallest learned relation needed to make that reduced equation executable?

The gradient closure can also be normalized in ways adapted to the moment variables. Numerical comparisons in the series show that predicting the derivative relation can outperform learning the high moment pointwise and can improve over conventional low-order moment closure in the tested transport regimes.

9.3 Radiative-transfer closure II: global hyperbolicity

A learned coefficient vector a_θ changes the principal matrix of the moment system. A generic network may produce a matrix with complex eigenvalues. The resulting PDE can lose hyperbolicity, so even a good offline regression fit may generate an ill-behaved time evolution.

The second paper seeks a symmetric positive-definite symmetrizer and derives restrictions under which the learned closure system is globally symmetrizable hyperbolic (Huang et al., 2021c). The construction is also arranged to retain dissipative structure and the correct diffusion limit as the Knudsen number tends to zero.

This is stronger than adding a loss term that penalizes complex eigenvalues on the training set. Hyperbolicity is encoded in the admissible parameterization of the closed PDE.

9.4 Radiative-transfer closure III: characteristic-speed parameterizations

A later construction exploits the lower-Hessenberg structure of the closure matrix. The network predicts spectral or polynomial information, and algebraic relations convert that information into closure coefficients (Huang et al., 2021b).

The work makes a useful tradeoff explicit:

- one architecture guarantees characteristic speeds in the physical interval but only weak hyperbolicity;
- another guarantees strict hyperbolicity but does not automatically bound all characteristic speeds by the physical speed limit.

This illustrates a general lesson in structure-preserving learning. Different desirable properties need not be simultaneously achievable through one simple parameterization. A mathematically serious model should state exactly which guarantee it provides.

9.5 Nonlinear BGK moment closure

The BGK kinetic model is nonlinear because the relaxation equilibrium depends on moments of f. Christlieb, Ding, Huang, and Krupansky extend the gradient-closure strategy to a Grad moment expansion of BGK (Christlieb et al., 2025).

The network is trained on moment data from kinetic solutions and predicts coefficients in a relation of the form (47). Its output layers are designed to enforce

$$\boxed{\text{hyperbolicity}} \qquad \text{and} \qquad \boxed{\text{Galilean invariance}}.$$

Galilean invariance means that a uniform shift of inertial frame does not change the physical content of the closure. A generic fully connected network acting on raw moments would not satisfy this automatically.

The model is coupled to a path-conservative numerical treatment because the highest retained moment equation is not conservative under the gradient closure. This is a revealing detail: the learned closure and the numerical PDE method must be designed together. One cannot assess the network independently of the closed system in which it will operate.

A schematic training loss is

$$\mathcal{L}(\theta) = \sum_{\text{samples}} \left| \partial_x m_{N+1} - \sum_{j=0}^{N} a_{j,\theta}(m) \partial_x m_j \right|^2 . \tag{48}$$

The target is not a complete solution field. It is the unresolved differential relation.

9.6 Learning a collision operator from molecular dynamics

A later project moves further down the modeling hierarchy. At the particle level, molecular dynamics contains many-body interactions and correlations. At the kinetic level these effects are compressed into a collision operator

$$\partial_t f + v \cdot \nabla_x f = \mathcal{Q}(f). \tag{49}$$

Rather than selecting a classical Landau- or Boltzmann-type form and calibrating a few coefficients, the model learns a generalized anisotropic collision operator directly from molecular-dynamics data (Zhao et al., 2025).

The operator is placed in a generalized metriplectic form designed to preserve

$$\begin{aligned} \int \mathcal{Q}(f)\,\mathrm{d}v &= 0, && \text{mass,} \\ \int v\,\mathcal{Q}(f)\,\mathrm{d}v &= 0, && \text{momentum,} \\ \int |v|^2 \mathcal{Q}(f)\,\mathrm{d}v &= 0, && \text{energy,} \end{aligned}$$

together with nonnegative entropy production. The learning procedure uses a weak formulation so that it need not estimate a high-dimensional probability density pointwise.

This is again closure modeling. The unknown object is larger than a moment relation, but it is still one component of a retained kinetic equation.

9.7 Why this is not a classical PINN

A classical PINN for BGK might represent

$$f_\theta(x, v, t) \approx f(x, v, t)$$

and minimize

$$\left\| \partial_t f_\theta + v \partial_x f_\theta - \frac{1}{\tau}\left(f_M[f_\theta] - f_\theta\right) \right\|^2. \tag{50}$$

Christlieb's closure program instead retains the kinetic or moment modeling hierarchy and learns only the missing relation needed by a low-dimensional reduced system.

The difference is not cosmetic:

	Classical PINN	**Christlieb-style closure**
Learned object	Full solution f_θ or m_θ	Missing moment-gradient or collision relation
Equation use	Residual in training loss	Determines hierarchy, closure location, admissibility, and deployed PDE
Solver	Neural optimization produces solution	Conventional solver advances closed equations
Structure	Often soft unless specially designed	Hyperbolicity, invariance, or conservation built into closure class
New initial condition	Usually new optimization	Reuse closure and solve reduced PDE again

9.8 Why this is not ordinarily called a neural operator

Abstractly, every closure is an operator. But in current scientific-ML usage, a neural operator usually denotes a discretization-aware map between functions, such as

$$f_0(x,v) \longmapsto f(x,v,t)$$

or

$$m(x,0) \longmapsto m(x,t).$$

The Christlieb closure more typically learns a local or semilocal constitutive rule

$$(m, \partial_x m) \longmapsto \partial_x m_{N+1}.$$

Its role is to complete a PDE, not to replace the entire PDE solution map.

9.9 The strengths and limits of the program

The strongest feature is structural control. The retained variables, missing term, characteristic matrix, invariances, and numerical method are identified explicitly. The model is modular and interpretable as a closure.

The price is specialization. One must know

1. which variables to retain;
2. where the hierarchy fails to close;
3. what functional form is plausible;
4. which conditions make the resulting PDE admissible;
5. how to solve the closed system robustly.

Hard constraints prevent specified pathologies but do not guarantee extrapolation accuracy. A hyperbolic model can remain well posed while being quantitatively wrong outside the training regime.

10 Qi Tang's program: structure-preserving learned dynamics, reduction, and operators

Tang's work occupies several categories. Some projects learn vector fields, some learn invariant manifolds, some learn symplectic flow maps, and some are genuine neural operators. The common thread is that dynamical or geometric structure is encoded in the model rather than left entirely to a regression loss.

10.1 Stabilized neural ODEs

A neural ODE learns a finite-dimensional vector field

$$\dot{u} = F_\theta(u).$$

In spatiotemporal systems, high-wavenumber behavior can be poorly captured by a generic dense network even when short-time state errors are small. Tang and collaborators propose a stabilized architecture

$$\dot{u} = L_\theta u + N_\theta(u), \tag{51}$$

where a sparse linear convolutional component and a dense nonlinear component are learned separately (Linot et al., 2023).

The split gives the model a more transparent mechanism for representing dissipative or high-frequency dynamics. Tests on viscous Burgers and Kuramoto–Sivashinsky emphasize not only one-step fit but shocks, high-wavenumber spectra, robustness to perturbed initial conditions, and whether long trajectories remain on the correct attractor.

This is best classified as data-driven system identification:

$$\boxed{\text{trajectory data} \longmapsto \text{learned vector field} \longmapsto \text{ODE integration.}}$$

The original PDE may have generated the training trajectories, but the learning method need not evaluate its residual.

10.2 Fast-slow neural networks

Consider a singularly perturbed system

$$\dot{x} = g(x, y, \varepsilon), \qquad \varepsilon\dot{y} = f(x, y, \varepsilon), \qquad 0 < \varepsilon \ll 1. \tag{52}$$

Under dissipative fast dynamics, trajectories approach a slow manifold

$$y = h(x, \varepsilon).$$

The reduced dynamics are approximately

$$\dot{x} = g(x, h(x, \varepsilon), \varepsilon).$$

The closure problem is to identify the manifold and the dynamics on it.

The fast-slow neural network learns an invertible transformation into fast-slow coordinates and a structured vector field in a modified Fenichel normal form (Serino et al., 2025). Schematically,

$$z \xmapsto{H_\theta} (x, y), \qquad \dot{y} = A_\theta(x, y, \varepsilon)y, \qquad \dot{x} = \varepsilon g_\theta(x, y, \varepsilon), \tag{53}$$

with the fast dynamics parameterized so that the manifold $y = 0$ is invariant and attracting.

The architecture therefore imposes the existence of a trainable attracting slow manifold as a hard constraint. Restricting to that manifold yields a learned reduced model. The examples include a Grad moment system, two-scale Lorenz–96, and Abraham–Lorentz dynamics.

Closure by learned coordinates

A traditional closure chooses reduced variables first and then models their missing flux. A fast-slow network can instead learn coordinates in which the slow variables and the attracting closure manifold become explicit.

This places the method conceptually near inertial-manifold reduction, equation-free modeling, and Mori–Zwanzig ideas, rather than near a residual PINN.

10.3 Symplectic Hénon networks and Poincaré maps

Hamiltonian flow maps preserve a symplectic form. In canonical coordinates, a map Φ is symplectic when

$$(D\Phi)^T J(D\Phi) = J. \tag{54}$$

A generic neural map does not satisfy this identity, and small geometric violations can produce artificial damping, energy drift, or incorrect phase-space structure under repeated iteration.

Tang and collaborators use Hénon-network layers whose composition is exactly symplectic. In the toroidal magnetic-field application, the network learns a Poincaré map from observations of a conventional field-line-following algorithm (Burby et al., 2021). The architecture exactly preserves the relevant flux constraint while evaluating much faster than repeated field-line integration.

The network is learning

$$z_{n+1} = \Phi_\theta(z_n),$$

not a PDE solution. The primary input is map data, and the primary prior is symplecticity.

Later work constructs a symplectic gyroceptron for nearly periodic maps. The architecture is nearly periodic and symplectic and supports a discrete adiabatic invariant, improving long-time behavior by design (Duruisseaux et al., 2022).

Recent reduced-order Hamiltonian work combines a symplectic encoder-decoder with symplectic latent dynamics, so both dimension reduction and time evolution preserve the underlying geometric class (Chen et al., 2025).

One should distinguish symplecticity from exact conservation of the original Hamiltonian. A symplectic integrator or map preserves phase-space geometry and often exhibits excellent long-time energy behavior, but it need not keep the original Hamiltonian exactly constant at every step.

10.4 Energetic variational learning

Many dissipative systems possess an energy-dissipation law

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}[u] = -\mathcal{D}[u], \qquad \mathcal{D}[u] \geq 0. \tag{55}$$

Tang and collaborators use energetic variational structure to infer unknown laws from probability-density or particle data (Lu et al., 2026). The governing information enters through an energy-dissipation principle rather than only through a pointwise strong-form PDE residual.

This is an instructive intermediate case. The method is physics-informed, but the "physics" is a variational law that organizes the admissible dynamics. It need not be presented as a standard PINN loss for the full PDE solution.

10.5 Local-global neural operator for conservation laws

Tang's local-global neural operator is a genuine neural-operator project. It learns a one-step flow map

$$\mathcal{G}_{\Delta t} : u^n \longmapsto u^{n+1} \tag{56}$$

for hyperbolic conservation laws (Wang et al., 2026).

The architecture combines

$$\mathcal{G}_\theta = \mathcal{G}_\theta^{\mathrm{global}} + \mathcal{G}_\theta^{\mathrm{local}}.$$

A Fourier branch models large-scale, long-range, relatively smooth interactions; a local multiresolution branch targets shocks, contacts, and localized nonsmooth features. The training objective combines physical-space prediction error with a high-frequency spectral term. As of August 2026, this work is a recent preprint and should be evaluated as such.

LGNO is much closer to FNO than to a PINN. It is supervised on high-order numerical solution data and learns a discrete state-transition operator. Its physics enters through the chosen flow-map target, local-global decomposition, conservation-oriented parameterization, boundary treatment, and spectral objective rather than primarily through a continuous PDE residual.

10.6 A classification of Tang's projects

Project	**Learned object**	**Hard or structural prior**	**Best classification**
Stabilized neural ODE	Vector field $L_\theta u + N_\theta(u)$	Linear/nonlinear split adapted to scale behavior	Data-driven system identification
Fast-slow network	Coordinate map, full vector field, and slow manifold	Attracting invariant slow manifold	Dynamical closure and model reduction
HénonNet Poincaré map	Discrete flow or section map	Exact symplecticity/flux preservation	Structure-preserving map surrogate

Project	Learned object	Hard or structural prior	Best classification
Symplectic latent ROM	Encoder, decoder, and latent flow map	Symplectic full-to-latent geometry	Geometric reduced-order modeling
Energetic variational learning	Unknown potential, mobility, or dissipative law	Energy-dissipation formulation	Structure-aware model discovery
LGNO	Function-to-function time-step map	Local-global decomposition and conservation-law inductive bias	Neural operator

10.7 The common methodological pattern

Across these projects the workflow is approximately

1. identify the unknown, expensive, or unstable component;
2. derive the mathematical structure that a usable approximation should possess;
3. parameterize a neural class satisfying as much of that structure as possible;
4. learn the remaining degrees of freedom from high-fidelity data;
5. evaluate the result inside long-time dynamics, not only by offline regression error.

This is more specialized than writing down a universal residual loss, but it exposes failure modes and aligns the learned object with the eventual numerical computation.

11 Side-by-side comparison: PINNs, neural operators, closures, and structured dynamics

Feature	Classical PINN	Standard neural operator	Christlieb closure	Tang structured models
Learned object	One solution $u(x,t)$	Solution or transition operator	Missing flux, moment relation, or collision operator	Vector field, slow manifold, coordinate map, or flow map
Typical data	Collocation points, BC/IC, sparse observations	Many input-output function pairs	High-fidelity kinetic moment or particle data	Trajectories, map pairs, or numerical solution pairs

Feature	**Classical PINN**	**Standard neural operator**	**Christlieb closure**	**Tang structured models**
Equation required in training?	Usually yes	No, unless physics-informed	The known hierarchy is structurally essential	Varies; often no residual is evaluated
Conventional solver retained?	Usually not for the represented solution	Usually replaced at inference	Yes	Sometimes: ODE integration for vector fields; no for direct flow maps
Cross-instance reuse	Limited in classical form	Central purpose	Reuse closure across reduced solves	Reuse learned dynamics or map within training regime
Typical hard guarantee	Few unless specially added	Few unless specially added	Hyperbolicity, invariance, conservation, entropy structure	Symplecticity, attracting manifold, scale structure, or conservation
Main economy	Avoid dense solution data; useful for inverse problems	Amortize many forward solves	Replace only expensive unresolved physics	Stable or geometry-aware long-time surrogate
Main risk	Nonconvex optimization and residual imbalance	Distribution shift and rollout instability	Incorrect closure ansatz or extrapolation	Wrong structural class or accumulated dynamical error

11.1 The computational economics

A classical PINN solves, for each instance a,

$$\theta^\star(a) = \arg\min_\theta \mathcal{L}_{\mathrm{PINN}}(\theta; a). \tag{57}$$

The online solve is itself a potentially expensive optimization.

A neural operator solves an expensive population-level training problem

$$\theta^\star = \arg\min_\theta \mathbb{E}_{a\sim\mu} \left\| \mathcal{G}_\theta(a) - \mathcal{G}^\dagger(a) \right\|^2, \tag{58}$$

then evaluates $\mathcal{G}_{\theta^\star}(a)$ cheaply for new inputs drawn from a similar distribution.

A learned closure trains $\mathcal{C}_\theta$ offline, then still solves a reduced equation for each new instance:

$$\partial_t m + \nabla \cdot F(m, \mathcal{C}_\theta(m, \nabla m)) = S(m). \tag{59}$$

Inference is not one network call; it is a cheaper PDE solve than the original kinetic, molecular, or fully resolved model.

A learned vector field similarly requires numerical integration, while a learned flow map can be iterated directly. The latter is often faster but places more responsibility on the learned map's recursive stability.

11.2 Which approach is natural for which task?

11.2.1 PINNs

PINNs are especially natural when the main goal is an inverse problem: sparse or irregular observations, unknown physical coefficients, and a reasonably trusted governing equation. They can also be attractive when complete training-solution ensembles are unavailable.

11.2.2 Neural operators

Neural operators are natural when many related forward solves are required, a representative training ensemble can be generated, and very fast online evaluation matters. Typical settings include optimization, uncertainty quantification, control, and digital twins.

11.2.3 Learned closures

A closure approach is natural when the full model is known but expensive, a meaningful reduced model already exists, and the precise unresolved term can be isolated. It preserves a mature solver and often permits more direct enforcement of well-posedness and asymptotic structure.

11.2.4 Structure-preserving learned dynamics

A structured vector field or flow map is natural when long-time behavior is controlled by geometry, invariant manifolds, stiffness, or attractors, and when ordinary one-step regression produces unstable rollouts.

12 Do these methods require a PDE?

The answer is no. They all require assumptions and information, but a partial differential equation is only one possible source of structure.

Method	**Must know a PDE?**	**What is actually required?**
Classical PINN	Usually yes, more generally a governing equation	Residual that can be evaluated, plus initial/boundary constraints and possibly data
Standard neural operator	No	Paired input and output functions from simulations, experiments, or another process
PINO	Usually yes	Operator-learning setup plus a known equation used in the physics loss
Christlieb moment closure	Yes, structurally	Known kinetic/moment hierarchy, a specific closure location, and admissibility conditions
Stabilized neural ODE	No	Trajectory data and the assumption that a useful continuous-time vector field exists
Fast-slow network	No	A dynamical system with dissipative fast variables and an attracting slow manifold
Symplectic HénonNet	No	Map observations plus the assumption that the target map is symplectic or flux-preserving
Supervised PDE neural operator	Not during training	State-transition or input-solution pairs, often generated by a PDE solver
World model	No	Sequential observations, actions, and enough state or memory to predict decision-relevant futures

12.1 Three distinct ways a PDE can enter

The PDE is evaluated during training. For a PINN or PINO,

$$\mathcal{N}[u_\theta] \quad \text{or} \quad \mathcal{N}_a[\mathcal{G}_\theta(a)]$$

is explicitly computed. The equation is operationally required.

The PDE determines the architecture and deployed model. For a learned moment closure, the equation determines the moments, the missing term, the principal matrix, the hyperbolicity conditions, and the solver. Even if training uses only high-fidelity data, the PDE is structurally essential.

The PDE merely generated the data. A neural ODE or supervised neural operator can train only on arrays

$$u(t_0), u(t_1), \dots, u(t_N)$$

or pairs (a, u). Experimental observations could replace simulation data without changing the core training formulation.

12.2 No PDE does not mean no model assumption

A neural ODE assumes the observed state is approximately Markovian in continuous time:

$$\dot{z} = F(z).$$

A one-step operator assumes

$$z_{n+1} = \Phi(z_n)$$

contains enough information for prediction. A fast-slow network assumes an attracting manifold. A symplectic network assumes the relevant map lies in a symplectic class. A world model assumes its latent state summarizes the history relevant to future consequences.

If coarse variables omit important hidden state, exact dynamics may require

$$\dot{z}(t) = F(z(t)) + \int_0^t K(z(t-s), s)\, \mathrm{d}s + \eta(t). \tag{60}$$

A Markovian neural model is then itself a closure approximation, even if it never mentions the original PDE.

Better question

Instead of asking only, "Does this method use a PDE?", ask:

> What information about the governing dynamics is supplied to the learner: an equation, solution data, a closure location, a variational law, a geometric class, trajectories, actions, or observation history?

13 The high-level numerical-approximation view

The preceding discussion can be condensed without losing its conceptual content.

13.1 A PINN approximates a function obeying an equation

$$(x, t) \longmapsto u_\theta(x, t), \qquad \mathcal{N}[u_\theta] \approx 0.$$

The network is a nonlinear trial function analogous in role, though not in training, to a finite-element or spectral ansatz.

13.2 A neural operator approximates an input-output map

$$a(\cdot) \longmapsto \mathcal{G}_\theta(a)(\cdot).$$

The network approximates the behavior of a solver over a family of inputs.

13.3 A learned closure approximates a missing relation

$$\text{resolved variables} \longmapsto \text{unresolved stress, flux, moment, or collision effect.}$$

The learned object is inserted into retained equations.

13.4 A neural ODE approximates a differential law

$$z \longmapsto F_\theta(z), \qquad \dot{z} = F_\theta(z).$$

A numerical integrator then constructs trajectories.

13.5 A learned flow map approximates finite-time evolution

$$z_t \longmapsto z_{t+\Delta t}.$$

The map is recursively composed. Stability of composition is central.

13.6 A structure-preserving network approximates within a restricted class

$$\Phi_\theta \in \{\text{symplectic maps}\}, \qquad A_\theta \in \{\text{hyperbolic systems}\}, \qquad \mathcal{M}_\theta \in \{\text{attracting manifolds}\}.$$

The approximation family is narrowed so that every candidate respects a chosen mathematical property.

The simplest accurate summary

The neural network is playing the role of a numerical approximation: sometimes of a function that obeys an equation, sometimes of a map that takes input to output, sometimes of a missing term inside an equation, and sometimes of the coordinates or geometry in which the dynamics become simple.

14 World models

14.1 The internal-simulator viewpoint

A world model extends the same approximation idea to an agent interacting with an environment. At its simplest, the physical or abstract state x_t evolves under an action a_t:

$$x_{t+1} = F(x_t, a_t). \tag{61}$$

A learned world model approximates

$$\widehat{x}_{t+1} = F_\theta(\widehat{x}_t, a_t). \tag{62}$$

Repeated application generates hypothetical futures:

$$\begin{aligned}
\widehat{x}_{t+1} &= F_\theta(x_t, a_t), \\
\widehat{x}_{t+2} &= F_\theta(\widehat{x}_{t+1}, a_{t+1}), \\
&\vdots \\
\widehat{x}_{t+H} &= F_\theta(\widehat{x}_{t+H-1}, a_{t+H-1}).
\end{aligned}$$

The agent can compare alternative action sequences before acting in the real environment.

World model at gross level

A world model is a learned numerical model of a controlled dynamical system, packaged for state estimation, counterfactual prediction, planning, or policy learning.

14.2 The state is usually hidden

An agent typically observes

$$o_t = H(x_t)$$

rather than the complete state. A single image may reveal position but not velocity; a sensor may omit important environmental variables. The model therefore constructs a latent state from history:

$$z_t = E_\theta(o_0, a_0, o_1, a_1, \ldots, o_t). \tag{63}$$

A typical latent world model contains

$$z_{t+1} \sim P_\theta(\cdot \mid z_t, a_t), \tag{64}$$

$$\widehat{o}_t \sim D_\theta(\cdot \mid z_t), \tag{65}$$

$$\widehat{r}_t = R_\theta(z_t, a_t), \tag{66}$$

where E_θ is an encoder or state estimator, P_θ is latent dynamics, D_θ is an observation decoder, and R_θ predicts reward, cost, danger, or another decision-relevant quantity.

The transition may be probabilistic because the environment is stochastic or because the latent state is incomplete.

14.3 What makes it a world model?

The term describes the *role* of the model more than one fixed architecture. A world model is expected to support some combination of

$$\boxed{\begin{array}{c}\text{state estimation} + \text{action-conditioned prediction}\\ +\text{counterfactual rollout} + \text{planning or policy learning}\end{array}}$$

A passive predictor learns

$$o_t \longmapsto o_{t+1}.$$

An interactive model learns

$$(o_t, a_t) \longmapsto o_{t+1}$$

or the corresponding latent transition. The action input makes alternative futures available.

A world model can be implemented by a recurrent network, transformer, neural ODE, neural operator, diffusion or video model, state-space model, conventional simulator, or hybrid combination. The term says what the model does for an agent.

14.4 Planning with a world model

Suppose the latent dynamics and cost are

$$z_{t+1} = F_\theta(z_t, a_t), \qquad c_t = C_\theta(z_t, a_t).$$

For a candidate action sequence $a_t, \ldots, a_{t+H-1}$, the model predicts

$$\widehat{J} = \sum_{k=0}^{H-1} \gamma^k C_\theta(\widehat{z}_{t+k}, a_{t+k}). \tag{67}$$

A model-predictive controller chooses

$$(a_t^\star, \ldots, a_{t+H-1}^\star) \in \arg\min \widehat{J}, \tag{68}$$

executes the first action, observes the real world again, updates the latent state, and replans.

Alternatively, imagined rollouts train a policy $\pi_\phi(a \mid z)$ and a value function. The Dreamer family follows this broad pattern, learning latent dynamics and improving behavior through imagined trajectories (Hafner et al., 2023).

The early *World Models* work learned a compressed visual representation and recurrent dynamics, then trained a compact controller partly inside generated trajectories (Ha and Schmidhuber, 2018).

14.5 A world model need not reconstruct everything

A model used for decision-making need not reproduce every pixel or microscopic detail. It may retain only information needed for reward prediction, value estimation, or planning.

MuZero is a central example. It learns a representation, dynamics, and prediction model that recursively produces rewards, policy information, and values useful for tree search without reconstructing the full observed environment or being given the environment rules (Schrittwieser et al., 2020).

This is analogous to closure modeling. A closure does not reproduce every molecular trajectory; it preserves the unresolved influence needed by the coarse equations. A task-oriented world model need not reproduce the whole world; it needs a state and dynamics sufficient for the decisions under consideration.

14.6 Generative interactive environments

The phrase "world model" has expanded to include action-conditioned generative simulators. A model may approximate

$$P_\theta\left(o_{t+1:t+H} \mid o_{\le t}, a_{t:t+H-1}\right), \tag{69}$$

the distribution of future observations conditioned on history and proposed actions.

Genie combines a spatiotemporal tokenizer, an autoregressive dynamics model, and a learned latent-action model to generate controllable interactive visual environments from video data (Bruce et al., 2024). Such systems are much larger and less explicitly physical than a PDE time-step operator, but mathematically they remain approximations to action-conditioned transition operators or probability kernels.

14.7 A neural operator can function as a world model

Consider a controlled fluid

$$\partial_t u = \mathcal{N}(u, a), \tag{70}$$

where a is a boundary control, body force, valve setting, or other intervention. The exact finite-time flow operator is

$$\Phi_{\Delta t} : (u_t, a_t) \longmapsto u_{t+\Delta t}. \tag{71}$$

A neural operator may learn

$$\Phi_{\theta,\Delta t}(u_t, a_t) \approx \Phi_{\Delta t}(u_t, a_t).$$

Used only for fast simulation, it is a neural time-step operator. Used by an agent to compare alternative control sequences and choose actions, it is the dynamics component of a world model.

Thus the categories overlap:

$$\boxed{\text{A neural operator can be the transition model inside a world model.}}$$

Likewise, a Christlieb-style closure plus a moment solver and an observation model could form a world model for a controlled kinetic system.

14.8 World models and closure models

The analogy is deeper than terminology.

	Closure model	**World model**
Retained representation	Coarse variables or moments	Latent state or belief state
Discarded information	Fine scales, high moments, particles	Unobserved world state and irrelevant detail
Learned object	Missing effect on retained equations	Transition and consequence model
Purpose	Predict coarse dynamics	Predict action-conditioned futures and support decisions
Exact complication	Memory and stochastic orthogonal dynamics	Partial observability, memory, uncertainty, and policy-dependent data
Typical simplification	Local deterministic constitutive law	Markovian latent transition

Both seek a representation that is smaller than the full underlying system but sufficient for a specified predictive task.

14.9 The controlled transition-operator viewpoint

From a numerical-analysis perspective, the fundamental object is often

$$\mathcal{P}_{\Delta t} : (z, a) \longmapsto \text{distribution of } z'. \tag{72}$$

For deterministic dynamics this reduces to

$$z' = F_{\Delta t}(z, a).$$

For stochastic or partially observed systems, the correct object is a Markov kernel

$$P(\,\mathrm{d}z' \mid z, a).$$

The world model approximates this kernel and composes it repeatedly.

This is closely related to system identification, state-space modeling, data assimilation, surrogate simulation, reduced-order modeling, and model-predictive control. The distinctive AI framing emphasizes that the learned model is used internally by an agent to imagine and evaluate consequences.

14.10 Why world modeling is harder than ordinary supervised approximation

14.10.1 Compounding error

A small one-step error does not imply a small rollout error:

$$F_\theta(z, a) \approx F(z, a) \quad \not\Rightarrow \quad F_\theta^H(z) \approx F^H(z).$$

This is the learned analogue of consistency and stability in numerical time integration.

14.10.2 Distribution shift under planning

The model is trained on actions that were taken. A planner evaluates actions that may be very different. It can push the model into regions with little training support and may even exploit model errors rather than discover genuinely good actions.

14.10.3 Hidden state and memory

If z_t does not contain enough information, no transition

$$z_{t+1} = F(z_t, a_t)$$

can be exact. One may need recurrence, a richer belief state, a window of past observations, latent stochastic variables, or an explicit memory kernel.

14.10.4 Multiple possible futures

A deterministic average may be meaningless when futures branch. The model should represent

$$z_{t+1} \sim P_\theta(\cdot \mid z_t, a_t)$$

or a distribution over future observation sequences.

14.10.5 Task relevance versus physical fidelity

A task-oriented latent model can make good decisions while representing the physical world incompletely. Conversely, a visually convincing video model may fail to preserve the causal consequences of actions. Prediction quality must be judged relative to the model's intended use.

14.11 The grossest possible world-model summary

One-line definition

A world model is a learned numerical approximation of enough state, dynamics, observations, and action consequences to act as an internal simulator for an agent.

In the same notation used throughout this chapter,

$$\begin{aligned} \text{PINN:}\quad & (x,t) \longmapsto u(x,t), \\ \text{neural operator:}\quad & \text{problem data} \longmapsto \text{solution}, \\ \text{closure:}\quad & \text{resolved variables} \longmapsto \text{missing term}, \\ \text{world model:}\quad & (\text{latent state}, \text{action}) \longmapsto (\text{next latent state}, \text{predicted consequences}). \end{aligned} \tag{73}$$

The additional ingredient is not a fundamentally new kind of approximation. It is recursive, action-conditioned use of the approximation for imagination, planning, and control.

15 A single conceptual ladder

The methods discussed here can be organized by how much of the mathematical model the network replaces.

1. **Approximate one coefficient or constitutive scalar.**

$$\mu_{\text{eff}} = \mu_\theta(\text{local state}).$$

2. **Approximate one closure tensor or missing flux.**

$$\tau = \mathcal{C}_\theta(\overline{u}, \nabla\overline{u}, \ldots).$$

3. **Approximate a reduced vector field.**

$$\dot{z} = F_\theta(z).$$

4. **Approximate a finite-time map.**

$$z_{n+1} = \Phi_\theta(z_n).$$

5. **Approximate a solution operator over a family.**

$$a(\cdot) \longmapsto \mathcal{G}_\theta(a)(\cdot).$$

6. **Approximate a latent controlled simulator.**

$$(z_t, a_t) \longmapsto P_\theta(\,\mathrm{d}z_{t+1},\,\mathrm{d}r_t,\,\mathrm{d}o_{t+1} \mid z_t, a_t).$$

Moving down this list generally increases the fraction of the simulator delegated to the learned model. It may increase speed and flexibility, but it also makes it harder to retain classical guarantees and to diagnose errors.

16 Practical questions for reading a scientific-ML paper

When confronted with a new "physics-informed," "operator-learning," "closure," or "world-model" paper, the following questions usually reveal what is actually happening.

16.1 What is the target object?

Is the network approximating

$$u, \quad \mathcal{G}, \quad \mathcal{C}, \quad F, \quad \Phi, \quad h, \quad \text{or} \quad P(\,\mathrm{d}z' \mid z, a)?$$

This is the first and most important classification.

16.2 Where does the supervision come from?

Possible answers include

- a strong-form differential-equation residual;
- a weak or variational identity;
- simulated input-output pairs;
- trajectories or state-transition pairs;
- high-fidelity closure targets;
- particle data;
- experimental observations;
- rewards, values, or planning targets.

16.3 What structure is hard, and what structure is soft?

Does the architecture guarantee conservation, hyperbolicity, positivity, symplecticity, invariance, or an attracting manifold? Or are these properties merely penalized on training samples?

16.4 What remains of the conventional solver?

Does the network replace the complete solve? Does it provide one flux inside a finite-volume method? Does an ODE solver integrate a learned right-hand side? Is a learned map simply iterated?

16.5 What is the relevant notion of generalization?

Generalization may mean a new initial condition, coefficient field, Knudsen number, forcing, geometry, action policy, time horizon, mesh, or physical regime. These are not interchangeable claims.

16.6 Is the model tested in closed loop?

For a closure, one wants an a posteriori PDE test. For a learned vector field, one wants long-time attractor and spectral tests. For a flow map, one wants recursive rollout. For a world model, one wants planning or policy performance under actions that may differ from the data-collection policy.

16.7 What failure is ruled out mathematically?

A useful guarantee is specific:

- eigenvalues remain real;
- covariance remains positive semidefinite;
- mass is conserved;
- a symplectic form is preserved;
- an invariant manifold exists;
- entropy production is nonnegative.

It should not be confused with a universal accuracy guarantee.

17 Final synthesis

Closure theory, PINNs, neural operators, structured learned dynamics, and world models belong to one broad landscape of numerical approximation, but they occupy different locations in a computational pipeline.

Closure theory

Closure begins when information is discarded. The reduced equations retain the effects of discarded scales through stresses, fluxes, correlations, memory, or stochastic terms. A closure supplies a tractable approximation to those effects.

PINNs

A classical PINN uses a neural network as a trial function for one solution and uses the governing equation primarily through a residual-based optimization problem. It normally requires a known governing equation, although that equation need not literally be a PDE.

Neural operators

A neural operator approximates a map between functions and amortizes a family of solves. It does not inherently require knowledge of a PDE; paired function data are enough. A PINO adds equation constraints to operator learning.

Christlieb

Christlieb's characteristic program learns missing kinetic closures inside retained moment or kinetic equations. The governing hierarchy identifies what is missing, while architecture and algebra enforce properties such as hyperbolicity, Galilean invariance, conservation, or entropy production.

Tang

Tang's work spans learned vector fields, attracting slow manifolds, symplectic maps, energetic variational laws, and neural operators. Its unifying feature is structure-preserving learning for reliable dynamics rather than one universal PINN recipe.

World models

A world model is an internal simulator for an agent. It approximates latent state evolution and action consequences, then uses recursive counterfactual rollouts for planning or policy learning. A neural operator, learned closure plus solver, or structured flow map can serve as a component of a world model.

At the highest level, all of these methods use neural networks as numerical approximation devices. The intellectually important distinctions are the identity of the approximated object, the source of supervision, the mathematical structure imposed on the approximation, and the way the learned object is composed with the rest of the model.

A Compact glossary

Closure
: A model for unresolved terms that appear after averaging, filtering, taking moments, projecting modes, or otherwise discarding information.

Constitutive relation
: A relation specifying material response, such as stress versus strain rate or heat flux versus temperature gradient. It is a form of closure.

RANS
: Reynolds-averaged Navier–Stokes; a mean-flow model in which all turbulent fluctuations are represented through modeled stresses and fluxes.

LES
: Large-eddy simulation; a filtered model that resolves large eddies and closes transfer across a selected spatial scale.

Moment closure
A relation expressing an unresolved high-order moment or its flux in terms of retained lower moments.

Mori–Zwanzig reduction
A projection formalism showing that exact reduced dynamics generally contain an instantaneous term, memory, and fluctuating or orthogonal dynamics.

PINN
In the narrow classical sense, a coordinate neural network trained as a solution ansatz using differential-equation residuals, initial and boundary conditions, and possibly observations.

Neural operator
A parameterized map between function spaces, usually trained to approximate a family of solution or transition operators.

PINO
A physics-informed neural operator that combines operator learning with equation-based constraints.

Neural ODE
A model in which a neural network parameterizes the right-hand side of an ODE and a numerical integrator constructs trajectories.

Flow map
A finite-time state-transition map $\Phi_{\Delta t} : z_t \mapsto z_{t+\Delta t}$.

Hard constraint
A property guaranteed for every model in the parameterized class, rather than encouraged by a finite penalty term.

World model
A learned representation of state, dynamics, observations, and action consequences used as an internal simulator for prediction, planning, or policy learning.

Latent state
An internal representation intended to summarize the observation history relevant to future prediction or control.

Transition kernel
A conditional probability law $P(\,\mathrm{d}z' \mid z, a)$ for the next state given current state and action.

B One-page comparison table

Paradigm	**Approximated object**	**Equation knowledge**	**What remains numerical?**	**Characteristic question**
RANS/LES closure	Stress or flux	Known resolved PDE	Full reduced PDE solve	How do discarded scales act on resolved scales?
Kinetic moment closure	High moment or gradient relation	Known hierarchy	Closed moment PDE solve	How can a finite moment set remain predictive and well posed?
Classical PINN	One solution function	Residual normally known	Nonlinear optimization over network parameters	Can one neural trial function fit equation and observations?
Neural operator	Function-to-function map	Optional	Mostly one learned inference call	Can one model amortize a family of solves?
PINO	Function-to-function map	Known residual	Operator inference, possibly fine-tuning	Can operator learning combine data and equation supervision?
Neural ODE	Vector field	Optional	ODE integration	Can trajectories identify a stable continuous-time law?
Symplectic map network	Finite-time map	No PDE required	Recursive map evaluation	Can long-time geometry be preserved exactly?
Fast-slow network	Coordinates, vector field, manifold	No PDE required	Full or reduced ODE integration	Can an attracting closure manifold be learned by design?
World model	Latent controlled transition kernel	No PDE required	Rollout, planning, policy optimization	Can an agent imagine useful action-conditioned futures?

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